

# EW2 Project 1

## Audio Amplifier Design Report

Team 10

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# 1 Introduction

## 1.1 Project Overview

The objective of this project is to design a multi-stage audio amplifier capable of amplifying weak audio signals to a level suitable for driving a loudspeaker. The design must operate within the audio frequency band of 20 Hz to 20 kHz, provide sufficient voltage and power gain, and maintain low distortion.

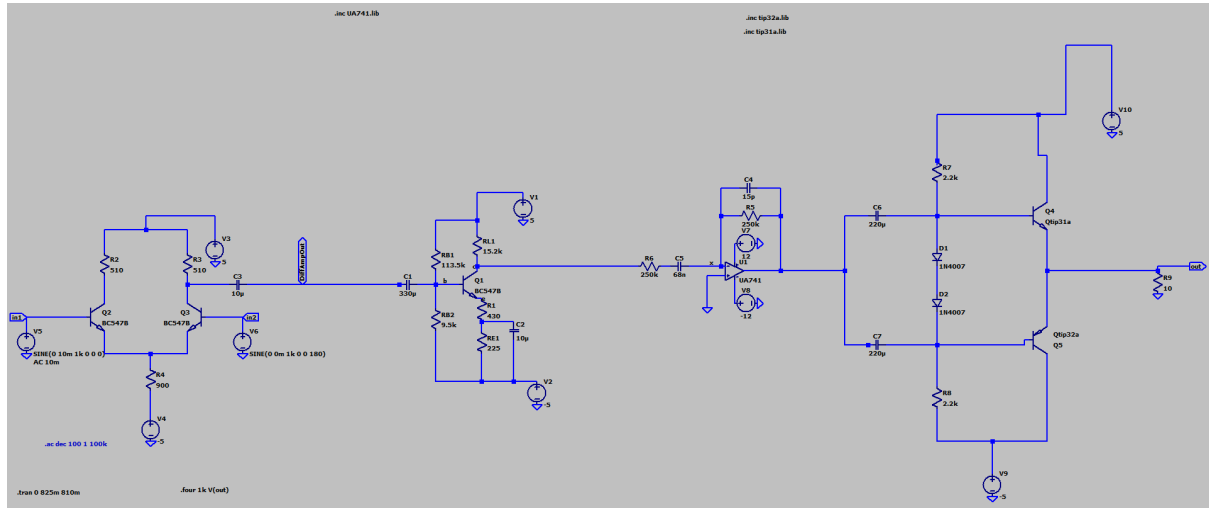


Figure 1: LTSpice Schematic of Audio Amplifier

## 1.2 Amplifier Architecture

The amplifier comprises four cascaded stages:

1. **Differential Amplifier (Pre-Amplifier):** Provides initial voltage gain with high common-mode rejection.
2. **Common-Emitter (CE) Gain Stage:** Provides the bulk of the voltage gain.
3. **Active Band-Pass Filter:** Limits the signal bandwidth to the audio range (20 Hz - 20 kHz) with unity or sub-unity gain.
4. **Class AB Power Amplifier:** Provides current gain (power gain) to drive a low-impedance load (speaker) at unity voltage gain.

## 1.3 Supply Voltage

The circuit operates with a dual-rail supply of  $V_{CC} = +5\text{ V}$  and  $V_{EE} = -5\text{ V}$ .

## 1.4 Hardware Implementation of Amplifier Circuit

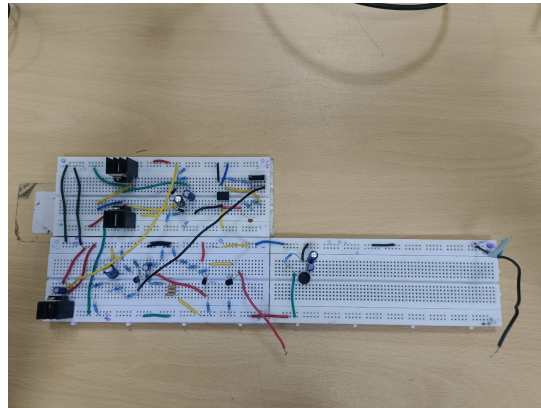


Figure 2: Hardware Implementation of Amplifier Circuit.

## 2 Microphone Input Circuit

The input signal for the amplifier is obtained from an **electric condenser microphone**. Such microphones contain a small internal JFET that requires a DC bias current for proper operation. Therefore, a simple biasing circuit is used to provide this current while allowing the generated AC audio signal to be extracted.

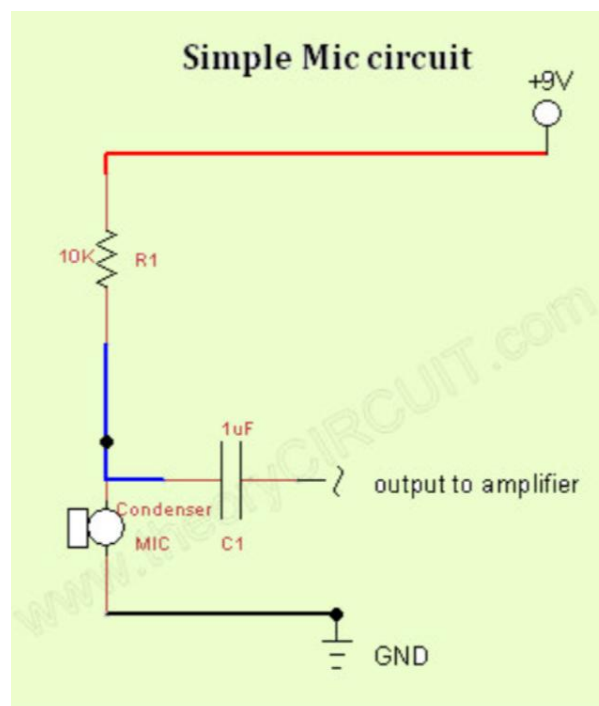


Figure 3: Electric microphone biasing circuit

The circuit consists of a bias resistor  $R_1$  ( $=10\text{k}\Omega$  in hardware), a coupling capacitor  $C_1$  ( $=57\mu\text{F}$  in hardware), and the microphone. The resistor  $R_1$  connects the microphone to the supply voltage and provides the necessary operating current for the internal JFET to operate in active region.

### 3 Differential Amplifier (Pre-Amplifier Stage)

#### 3.1 Theory and Circuit Description

The BJT differential pair is one of the most fundamental building blocks in analog circuit design. It consists of two matched BJT transistors whose emitters are connected together through a common tail resistor  $R_E$  (or a current source). The circuit amplifies the *difference* between two input signals while rejecting signals common to both inputs.

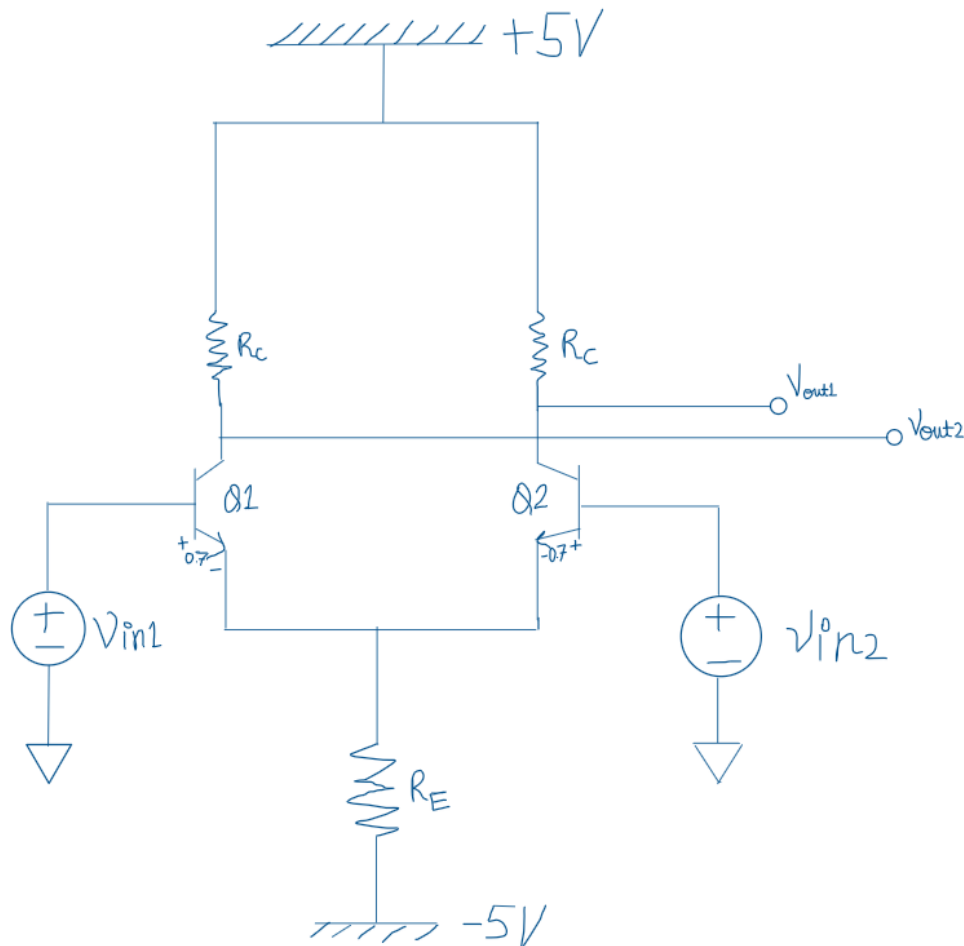


Figure 4: BJT Differential Pair circuit schematic.

#### Pre-Amplifier Stage

- The input signal from the microphone cannot be directly fed to the common-emitter or power amplifier because it is very small and susceptible to noise.
- A pre-amplifier is used to increase the signal level to a suitable value (line level) before further amplification.
- The pre-amplifier should have:
  - High input impedance to avoid loading the input source.
  - Low output impedance to prevent voltage drop at the output.

- In this design, the pre-amplifier stage is implemented using a **differential amplifier** with a single-ended (unbalanced) output configuration, where one input is connected to the microphone and the other is grounded.

## 3.2 DC Bias Analysis

### 3.2.1 Assumptions

We assume that the base current is negligible ( $I_B \approx 0$  A), which is valid for high- $\beta$  transistors. Since both transistors are matched:

$$I_{C1} = I_{C2}, \quad I_{E1} = I_{E2}$$

and  $I_C \approx I_E$  (since  $\beta \gg 1$ ).

### 3.2.2 KVL Analysis

Applying KVL from the base of BJT 1 through the emitter to  $V_{EE}$ :

$$-0.7 - I_E R_E = -5$$

$$I_E R_E = 4.3$$

$$\boxed{I_E = \frac{4.3}{R_E}} \tag{1}$$

Applying KCL at the emitter node:

$$I_{E1} + I_{E2} = I_E \tag{2}$$

Since both transistors are matched:  $I_{E1} = I_{E2} = \frac{I_E}{2}$ .

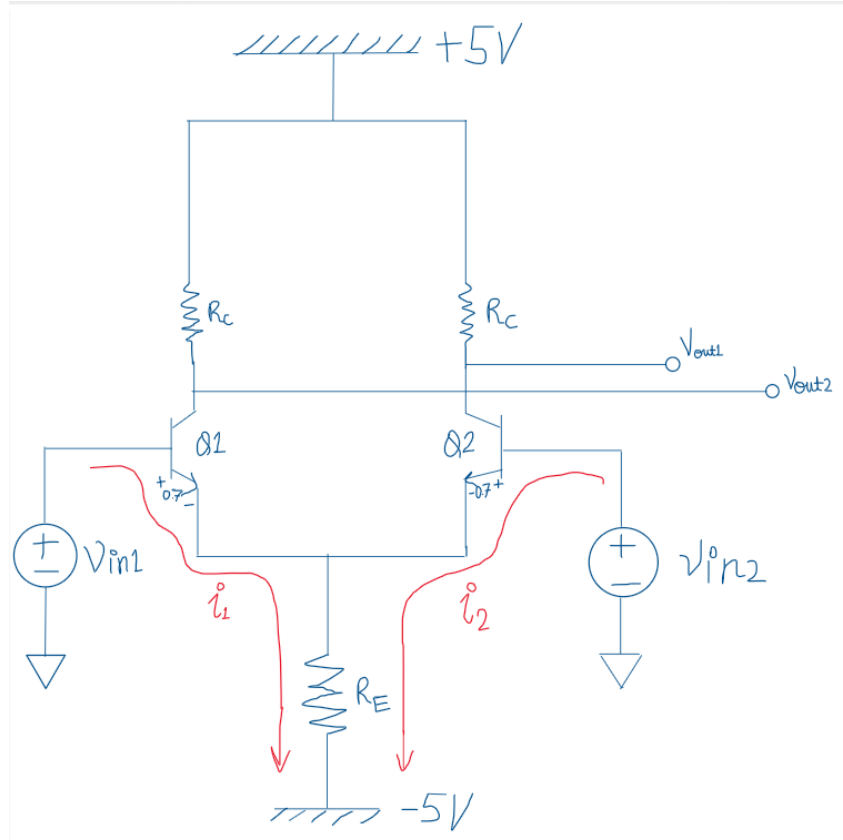


Figure 5: BJT Differential Pair circuit schematic.

### 3.3 Small-Signal AC Analysis and Gain Derivation

#### 3.3.1 Small-Signal Model

In the small-signal analysis, all DC sources are set to zero and the transistors are replaced by their hybrid- $\pi$  equivalent models. The key small-signal parameter is:

$$r_e = \frac{V_T}{I_E} = \frac{1}{g_m}$$

where  $V_T \approx 25 \text{ mV}$  at room temperature (300 K) and  $g_m$  is the transconductance.

#### 3.3.2 Derivation of Differential Gain

Applying KVL in BJT 1 and BJT 2 small-signal loops:

$$v_{in1} - i_{e1}r_e - (i_{e1} + i_{e2})R_E = 0 \quad (1)$$

$$v_{in2} - i_{e2}r_e - (i_{e1} + i_{e2})R_E = 0 \quad (2)$$

Rearranging into standard form:

$$i_{e1}(r_e + R_E) + i_{e2}R_E = v_{in1} \quad (3)$$

$$i_{e1}R_E + i_{e2}(r_e + R_E) = v_{in2} \quad (4)$$

In matrix form:

$$\begin{bmatrix} r_e + R_E & R_E \\ R_E & r_e + R_E \end{bmatrix} \begin{bmatrix} i_{e1} \\ i_{e2} \end{bmatrix} = \begin{bmatrix} v_{in1} \\ v_{in2} \end{bmatrix}$$

The determinant of the coefficient matrix is:

$$\Delta = (r_e + R_E)^2 - R_E^2 = r_e^2 + 2R_E r_e$$

By Cramer's rule:

$$i_{e1} = \frac{(R_E + r_e)v_{in1} - R_E v_{in2}}{r_e^2 + 2R_E r_e} \quad (3)$$

$$i_{e2} = \frac{(R_E + r_e)v_{in2} - R_E v_{in1}}{r_e^2 + 2R_E r_e} \quad (4)$$

The differential output voltage is:

$$v_{out} = v_{o2} - v_{o1} = (i_{e1} - i_{e2})R_C$$

Substituting equations (3) and (4):

$$\begin{aligned} v_{out} &= \left( \frac{(R_E + r_e)v_{in1} - R_E v_{in2}}{r_e^2 + 2R_E r_e} - \frac{(R_E + r_e)v_{in2} - R_E v_{in1}}{r_e^2 + 2R_E r_e} \right) R_C \\ &= \frac{(R_E + r_e)(v_{in1} - v_{in2}) + R_E(v_{in1} - v_{in2})}{r_e^2 + 2R_E r_e} \cdot R_C \\ &= \frac{(2R_E + r_e)(v_{in1} - v_{in2})}{r_e(r_e + 2R_E)} \cdot R_C \\ &= \frac{R_C}{r_e}(v_{in1} - v_{in2}) \end{aligned} \quad (5)$$

Since the differential input  $v_{in} = v_{in1} - v_{in2}$ :

$$A_d = \frac{v_{out}}{v_{in}} = -\frac{R_C}{r_e} = -g_m R_C \quad (6)$$

This is the **standard differential gain formula** for a BJT differential pair. The negative sign indicates signal inversion.

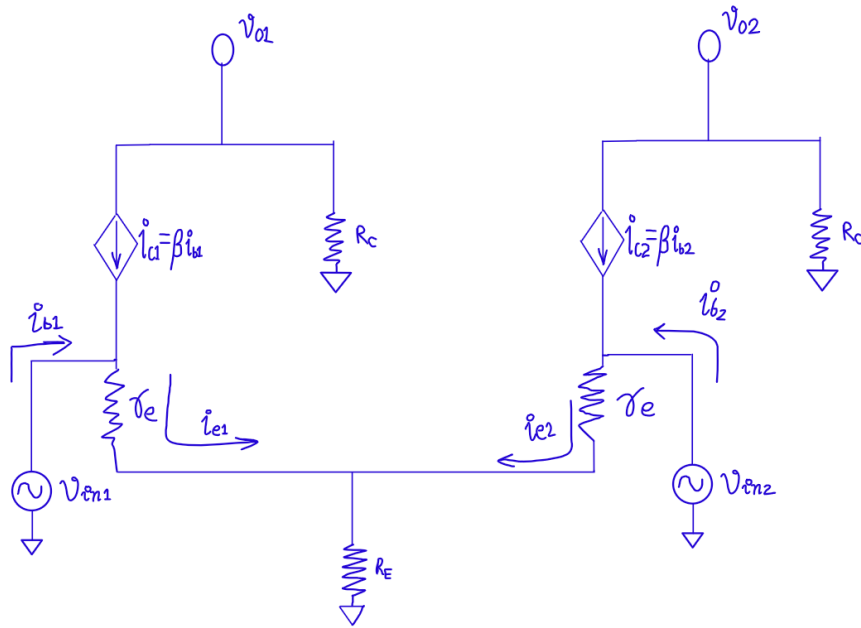


Figure 6: BJT Differential Pair circuit schematic.

### 3.4 Component Value Design

Setting the target differential gain  $|A_d| = 49$ :

$$g_m R_C = 49 \implies \frac{I_C R_C}{V_T} = 49$$

Since  $I_C \approx \frac{I_E}{2}$  and the tail current is

$$I_E = \frac{4.3}{R_E}$$

we have

$$I_C = \frac{4.3}{2R_E}$$

Substituting:

$$49 = \frac{I_C R_C}{V_T}$$

$$49 = \frac{4.3}{2V_T} \cdot \frac{R_C}{R_E}$$

$$\frac{R_C}{R_E} = \frac{49 \times 2V_T}{4.3} = \frac{49 \times 2 \times 0.025}{4.3} = \frac{2.45}{4.3} \approx 0.57$$

**Chosen component values:**

$$\boxed{R_E = 900 \Omega, \quad R_C = 510 \Omega} \quad (7)$$

**Verification:**

$$I_E = \frac{4.3}{900} = 4.78 \text{ mA}$$

$$I_{E1} = I_{E2} = \frac{4.78}{2} = 2.39 \text{ mA}$$

$$g_m = \frac{I_C}{V_T} = \frac{2.39 \times 10^{-3}}{25 \times 10^{-3}} = 0.0956 \text{ S}$$

Achieved differential gain:

$$|A_d| = g_m R_C$$

$$|A_d| = 0.0956 \times 510 \approx 48.8$$

$$|A_d| \approx 49$$

**Reason for Gain Reduction in Simulation:**

The calculated differential gain of the differential pair is

$$A_d = g_m R_C \approx 49$$

which represents the *differential gain* measured between the two collector outputs. However, the output is taken from only one collector (single-ended output). For a differential amplifier, the single-ended gain is approximately half of the differential gain:

$$A_v = \frac{A_d}{2}$$

Thus,

$$A_v \approx \frac{49}{2} \approx 24.5$$

which matches the simulated gain of approximately 24. This reduction occurs because the differential signal is shared between the two collector outputs.

### 3.5 Input and Output Impedance

#### 3.5.1 Input Impedance ( $R_{in}$ )

The input impedance looking into the base of either transistor is:

$$R_{in} = r_\pi = \frac{\beta}{g_m} = \frac{V_T \beta}{I_C}$$

**Standard derivation:** From the hybrid- $\pi$  model,  $r_\pi = \frac{v_{be}}{i_b}$ . Since  $i_c = g_m v_{be}$  and  $i_c = \beta i_b$ :

$$r_\pi = \frac{v_{be}}{i_b} = \frac{i_c / g_m}{i_c / \beta} = \frac{\beta}{g_m}$$

Using the previously calculated current:

$$I_C = 2.39 \text{ mA}$$

$$g_m = \frac{I_C}{V_T} = \frac{2.39 \times 10^{-3}}{25 \times 10^{-3}} = 0.0956 \text{ S}$$

Assuming  $\beta = 100$ :

$$R_{in} = r_\pi = \frac{\beta}{g_m}$$

$$R_{in} = \frac{100}{0.0956}$$

$$R_{in} \approx 1.05 \text{ k}\Omega$$

$$\boxed{R_{in} \approx 1.05 \text{ k}\Omega} \tag{8}$$

### 3.5.2 Output Impedance ( $R_{out}$ )

The output impedance is the parallel combination of the collector resistor and the output resistance of the transistor:

$$R_{out} = R_C \parallel r_o$$

where  $r_o$  models the Early voltage effect:

$$r_o = \frac{V_A}{I_C}$$

Assuming  $V_A = 100$  V:

$$r_o = \frac{100}{0.00239} \approx 41.8 \text{ k}\Omega$$

Since  $R_C \ll r_o$ :

$$R_{out} = (510 \parallel 41.8 \text{ k}) \Omega$$

$$R_{out} \approx 503 \Omega$$

$$\boxed{R_{out} \approx 510 \Omega} \tag{9}$$

## 3.6 Common-Mode Rejection Ratio (CMRR)

### 3.6.1 Theory

The CMRR quantifies the ability of a differential amplifier to reject common-mode signals relative to differential signals. It is defined as:

$$\text{CMRR} = \frac{|A_d|}{|A_c|}$$

or in decibels:

$$\text{CMRR (dB)} = 20 \log_{10} \left( \frac{|A_d|}{|A_c|} \right)$$

### 3.6.2 Common-Mode Gain and CMRR

For a common-mode input ( $v_{in1} = v_{in2} = v_{cm}$ ), both transistors conduct equally. Applying KVL for the common-mode half-circuit:

$$v_{cm} = i_e r_e + 2i_e R_E$$

$$i_e = \frac{v_{cm}}{r_e + 2R_E}$$

The common-mode output at each collector is

$$v_{o,cm} = -i_e R_C$$

Substituting  $i_e$ ,

$$v_{o,cm} = -\frac{R_C}{r_e + 2R_E}v_{cm}$$

Thus the common-mode gain is

$$A_c = \frac{v_{o,cm}}{v_{cm}} = -\frac{R_C}{r_e + 2R_E}$$

Since  $r_e \ll 2R_E$ , the expression simplifies to

$$A_c \approx -\frac{R_C}{2R_E}$$

Using the designed component values

$$R_C = 510 \Omega, \quad R_E = 900 \Omega$$

$$A_c \approx -\frac{510}{2 \times 900}$$

$$A_c \approx -0.283$$

**Theoretical common-mode gain:**

$$|A_c| \approx 0.28$$

From LTSpice AC analysis, the common-mode gain obtained is

$$A_c \approx 0.0027$$

Using this value,

$$CMRR = 20 \log_{10} \left( \frac{49}{0.0027} \right)$$

$$CMRR \approx 85 \text{ dB}$$

### Reason for Difference

The theoretical derivation uses simplified small-signal assumptions such as perfect transistor symmetry and the approximation  $r_e \ll 2R_E$ . However, a full transistor model is used which includes parameters such as transistor gain, Early effect, and internal resistances. These effects improve the common-mode rejection of the differential pair, resulting in a much smaller common-mode gain in simulation.

### 3.7 Simulation Validation

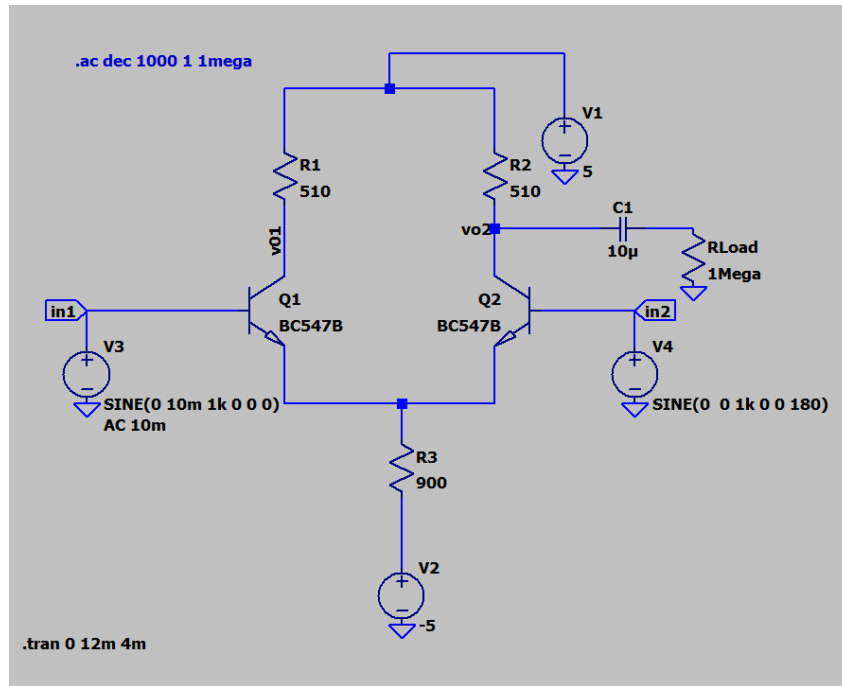


Figure 7: LTSpice Schematic of Differential Amplifier

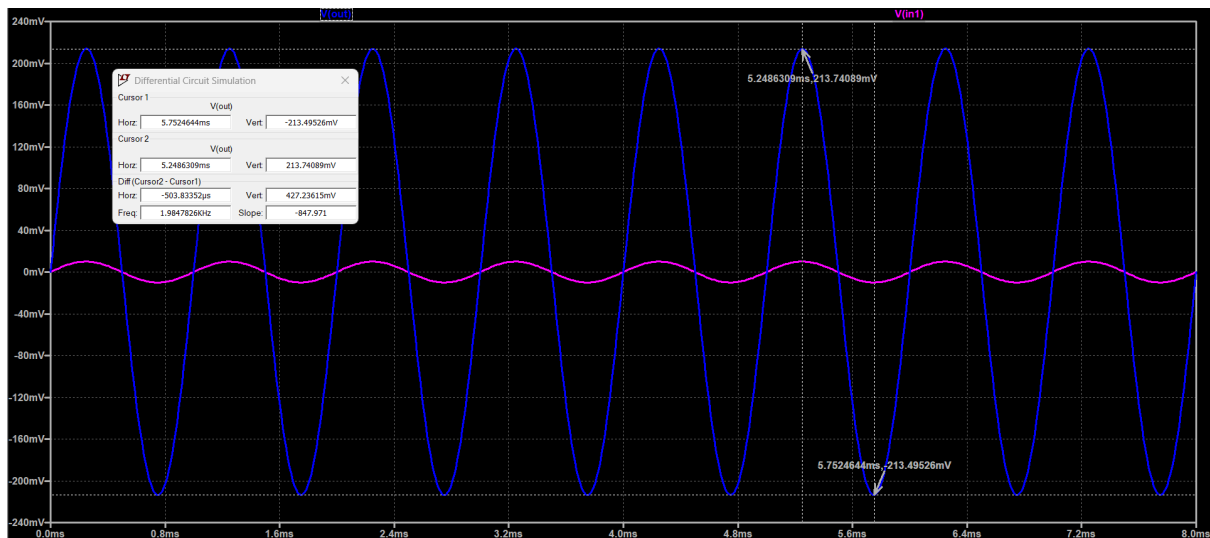


Figure 8: LTSpice Simulation

**Legend:** —  $V_{o2}$  —  $V_{in1}$

$$\text{Gain till Differential Amp stage, LTSpice} = \frac{427.23 V_{p-p}}{20 mV_{p-p}} = 21.35$$

\* The measured voltage gain (21.35) is lower than the expected value (24). This deviation occurs because the power amplifier stage draws a relatively large current, introducing a loading effect on the preceding stages. As a result, the effective gain of the differential amplifier stage is reduced.

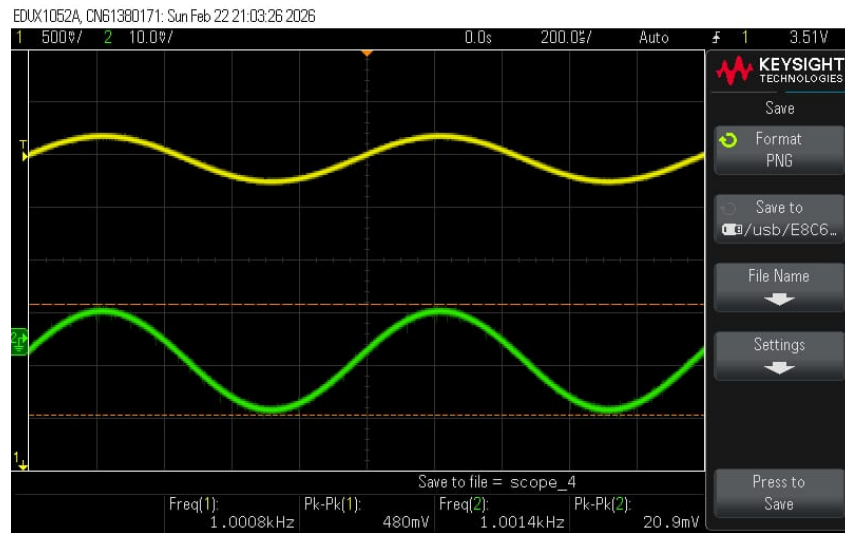


Figure 9: Scope output of hardware implementation: Differential Amplifier Output

**Legend:** — Input ( $V_{in1}$ ) — Output ( $V_{o2}$ )

$$\text{Gain till Differential Amp stage, Hardware} = \frac{480 V_{p-p}}{20 mV_{p-p}} = 24$$

## 4 Common-Emitter (CE) Gain Stage

### 4.1 Theory and Circuit Description

In our design, a partially-bypassed emitter resistor configuration of single-transistor CE amplifier is used.

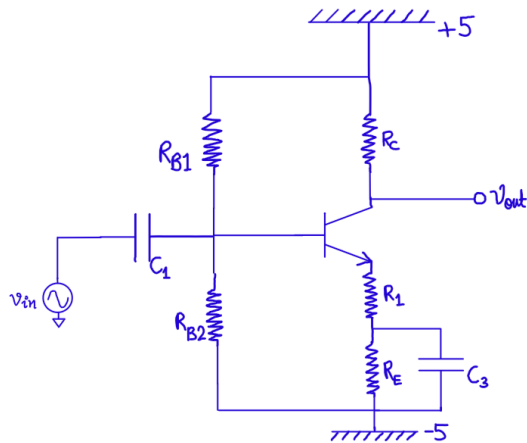


Figure 10: CE amplifier schematic

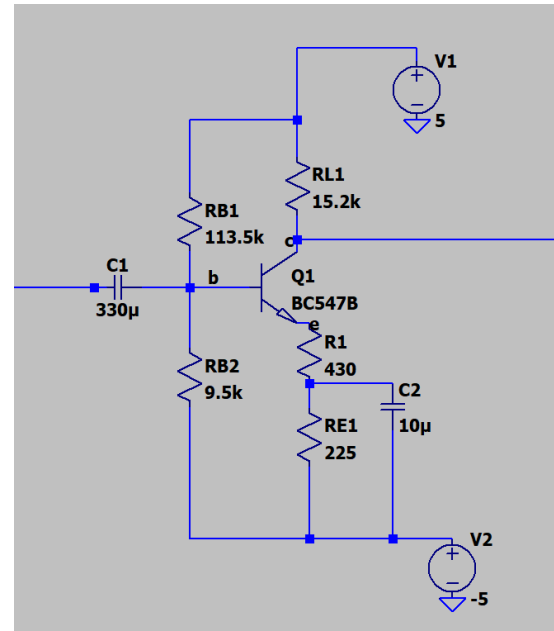


Figure 11: LTSpice Circuit

#### 4.1.1 Amplifier Configuration Choice

The CE configuration was chosen because:

- It provides the highest voltage gain among the three basic BJT configurations (CE, CB, CC).
- It provides moderately high input impedance, suitable for receiving the signal from the differential amplifier stage.

## 4.2 DC Bias Design (Voltage Divider Bias)

We have used a voltage divider bias for implementing the biasing network for the CE amplifier stage. It uses two resistors ( $R_{B1}$  and  $R_{B2}$ ) to set the DC base voltage independent of  $\beta$ . This significantly improves thermal stability as it creates a robust voltage divider, the quiescent base voltage  $V_B$  remains relatively constant even if the transistor's characteristics shift due to temperature variations.

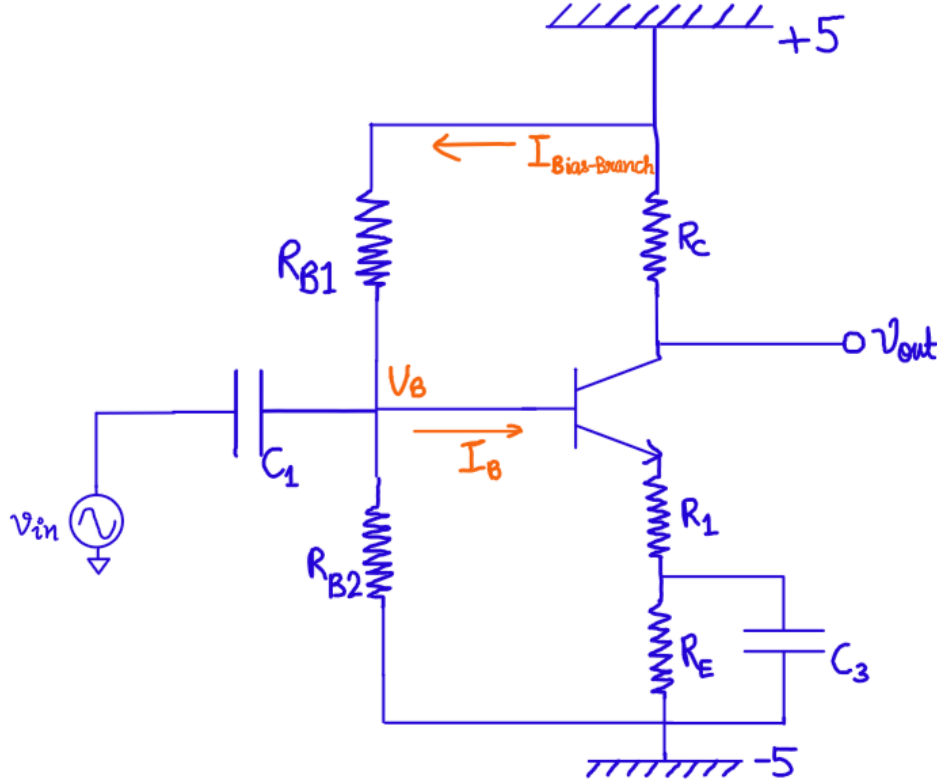


Figure 12: Biasing Network

### 4.2.1 Design Calculations

To achieve a perfectly symmetrical maximum swing, we bias the DC collector voltage ( $V_{CQ}$ ) exactly at the midpoint of the rails (0 V).

The output voltage of a CE amplifier is the collector voltage  $V_C$ . This voltage is physically constrained between two limits:

- **Upper Swing Limit:** If  $V_C$  attempts to rise above  $V_{CC}$ , the transistor enters cutoff. The output clips at  $V_{CC}$ .
- **Lower Swing Limit:**  $V_{C,\min} = V_E + V_{CE,\text{sat}}$ . If  $V_C$  drops too low and approaches  $V_E$ , the transistor enters saturation ( $V_{CE,\text{sat}} \approx 0.2 \text{ V}$ ).

By setting  $V_E = -4.9 \text{ V}$ , we push the lower swing limit as low as possible. This creates the largest possible range for  $V_C = V_{out}$  to swing, which maximizes output swing. To achieve the symmetrical maximum swing, the DC quiescent collector voltage  $V_{CQ}$  is biased at the midpoint of the output range. In our design,  $V_{CQ} \approx 0 \text{ V}$ .

Setting  $V_E = -4.9\text{ V}$  and  $V_C = 0\text{ V}$ , for BJT to be in active region, we make the biasing network such that it constrains  $V_B$  to be:

$$V_B = V_E + V_{BE} = -4.9 + 0.7 = -4.2\text{ V} \quad (10)$$

By KVL, assuming  $I_B \ll I_{\text{Bias-branch}}$ :

$$\frac{5 - V_B}{R_{B1}} = \frac{V_B + 5}{R_{B2}}$$

On Rearranging,

$$5(R_{B2} - R_{B1}) = V_B(R_{B1} + R_{B2})$$

Substituting  $V_B = -4.2\text{ V}$  and simplifying, we find the ratio of the biasing resistors:

$$\frac{R_{B1}}{R_{B2}} = \frac{23}{2} = 11.5 \quad (11)$$

We select a value for  $R_{B2}$  and  $R_{B1}$  which would be nearly, of the order of magnitude  $R_{B1} \approx 100\text{ k}\Omega$  and  $R_{B2} \approx 10\text{ k}\Omega$  so that the Input impedance of the CE Amplifier stage remains fairly above the output impedance of Differential Amplifier Stage ( $\approx 503\Omega$ ) to prevent Amplitude degradation due to loading, Thus, based on these considerations and iterative optimization we choose:

$$\boxed{R_{B1} \approx 113.5\text{ k}\Omega, \quad R_{B2} \approx 9.5\text{ k}\Omega} \quad (12)$$

Note that the ratio  $\frac{R_{B1}}{R_{B2}} \approx 11.94$ , which is not significantly different from the initially derived ratio.

#### 4.2.2 Bias Current Verification

The bias branch current through the voltage divider:

$$I_{\text{Bias-branch}} = \frac{10}{15 \times 10^3} = 86\text{ }\mu\text{A} \gg I_B$$

For worst-case BC547 transistor,  $\beta = 200$ :

$$I_B = \frac{I_C}{\beta} = \frac{1 \times 10^{-3}}{200} = 5\text{ }\mu\text{A}$$

Since  $I_B \ll I_{\text{Bias-branch}}$ , the assumption of negligible  $I_B$  in bias calculations was **consistent and valid**.

#### 4.2.3 Verification of Active Region of Operation

To ensure the BJT remains in the **Forward Active** region, the following two conditions need to be satisfied:

1. **Base-Emitter Junction (Forward Bias):**  $V_{BE} \geq V_{BE,\text{on}}$ , typically  $\approx 0.7\text{ V}$ .  
In our voltage divider network, this is maintained by Equation 10. Since we ensure  $V_B = -4.24\text{ V}$  and  $V_E = -4.94\text{ V}$  (exact value in LTSpice), we have:

$$V_{BE} = -4.24\text{ V} - (-4.94\text{ V}) = 0.7\text{ V}$$

## 2. Collector-Base Junction (Reverse Bias): $V_{CE} > V_{CE,sat}$ , ( $V_{CE,sat} \approx 0.2\text{ V}$ )

By biasing  $V_E$  at  $-4.94\text{ V}$ , the collector voltage  $V_C$  can swing as low as  $-4.74\text{ V}$  before the transistor enters saturation. Since our quiescent point  $V_{CQ}$  is biased at  $0\text{ V}$ , the transistor remains safely in the active region for the majority of the output signal range.

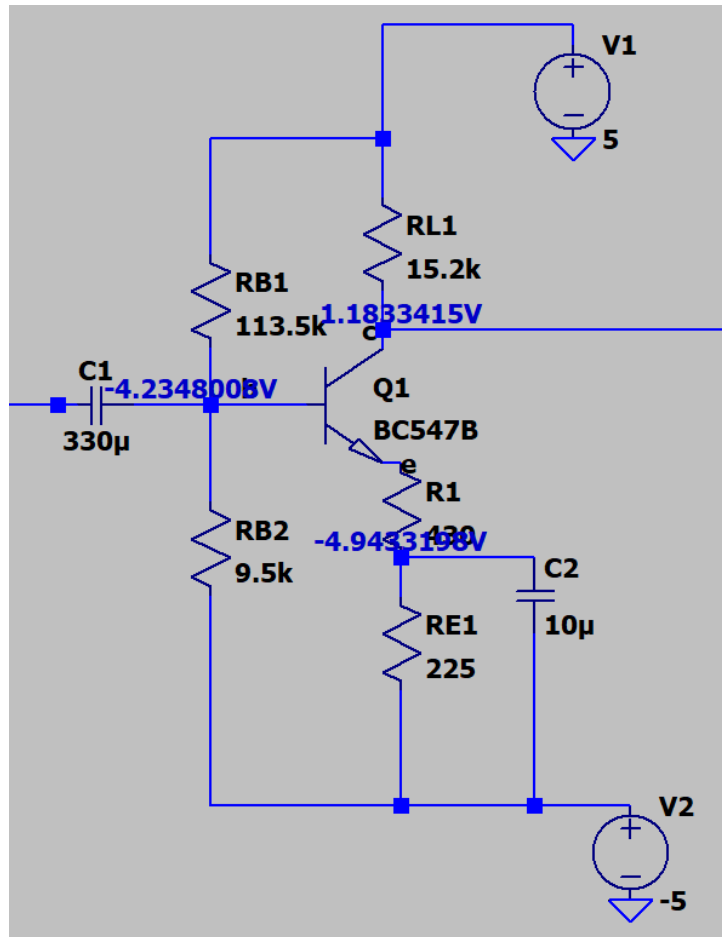


Figure 13: CE amplifier bias points

- **Base DC Voltage:**

$$V_{BQ} = -4.23\text{ V}$$

- **Emitter DC Voltage:**

$$V_{EQ} = -4.94\text{ V}$$

- **Collector DC Voltage:**

$$V_{CQ} = 1.184\text{ V} \quad V_{CE} = V_{CQ} - V_{EQ} = 6.124 > V_{CE,sat}$$

since  $V_{CE,sat} \approx 0.2\text{ V}$  and,  $V_{BE} = 0.71\text{ V} > 0.7\text{ V}$  Hence, we conclude that our transistor is in active region of operation (Maximum value of  $V_{out}$  is  $4.5\text{ V}$  above  $0\text{ V}$  so,  $V_C$  stays below  $5\text{ V}$  -> No clipping)

### 4.3 Small-Signal Gain Calculation

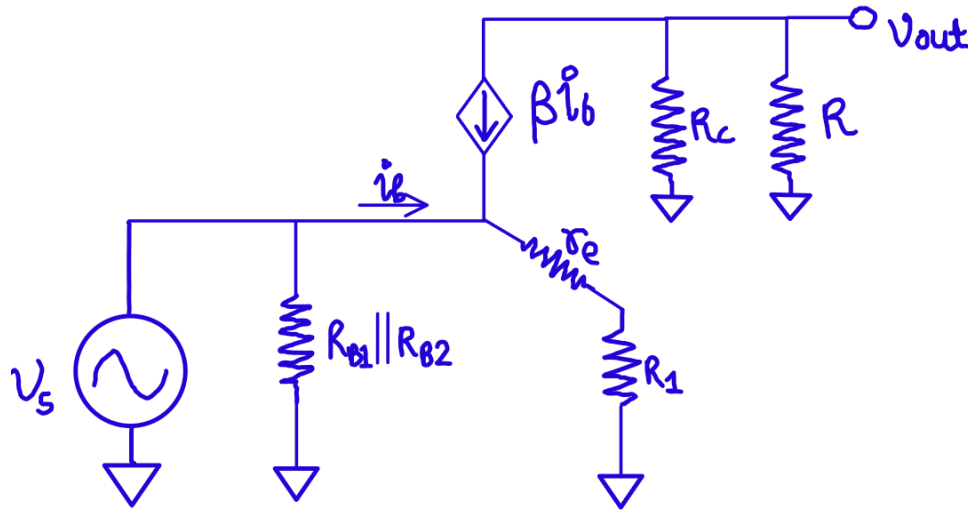


Figure 14: CE amplifier small-signal schematic.

#### 4.3.1 Gain Expression

Applying KVL:

$$v_s - (\beta + 1)i_B(r_e + R_1) = 0$$

$$v_s = (\beta + 1)i_B(r_e + R_1)$$

$$v_{out} = -\beta i_B(R_C \parallel R)$$

$$\frac{v_{out}}{v_s} = -\frac{\beta}{\beta + 1} \left( \frac{R_C \parallel R}{r_e + R_1} \right)$$

$$\boxed{A_v = -\frac{R_C \parallel R}{r_e + R_1}} \quad (13)$$

where  $r_e = V_T/I_E$  is the intrinsic small-signal emitter resistance. Now, we assume Load resistance  $R$  is very large (the input impedance of Bandpass filter)

Now, since  $V_C$  is biased at 0 V, by DC analysis:

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{5}{R_C}$$

$$A_v = -\frac{R_C}{r_e + R_1}$$

$$\boxed{A_v = -\frac{R_C}{\frac{V_T}{I_E} + R_1}} \quad (14)$$

### 4.3.2 Design Equations for $R_1$ and $R_E$

Setting  $I_E \approx I_C = 5/R_C$ , we derive the ratio  $R_1/R_C$ :

$$\begin{aligned}
 -\frac{R_C}{\frac{R_C V_T}{5} + R_1} &= A_v \\
 -\frac{1}{\frac{V_T}{5} + \frac{R_1}{R_C}} &= A_v \\
 \boxed{\frac{R_1}{R_C} = -\frac{1}{A_v} - \frac{V_T}{5} = \frac{1}{|A_v|} - \frac{V_T}{5}} & \quad (15)
 \end{aligned}$$

By DC analysis, we have current in Emitter branch  $I_E$ , (as capacitance  $C_3$  acts as open-circuit for DC signals):

$$I_E = \frac{5 + V_E}{R_E + R_1}$$

Now, for the ratio  $R_E/R_C$ :

$$\begin{aligned}
 |A_v| &= \frac{R_C}{R_1 + \frac{V_T(R_1 + R_E)}{5 + V_E}} \\
 \frac{1}{|A_v|} &= \frac{R_1}{R_C} \left( 1 + \frac{V_T}{5 + V_E} \right) + \frac{R_E}{R_C} \cdot \frac{V_T}{5 + V_E}
 \end{aligned}$$

Now, we put the ratio  $R_1/R_C$  found in eq. 15 to get:

$$\begin{aligned}
 \frac{1}{|A_v|} &= \left( \frac{1}{|A_v|} - \frac{V_T}{5} \right) \left( 1 + \frac{V_T}{5 + V_E} \right) + \frac{R_E}{R_C} \left( \frac{V_T}{5 + V_E} \right) \\
 \frac{1}{|A_v|} \left( \frac{5 + V_E}{V_T} \right) &= \left( \frac{1}{|A_v|} - \frac{V_T}{5} \right) \left( \frac{5 + V_E}{V_T} + 1 \right) + \frac{R_E}{R_C} \\
 \frac{R_E}{R_C} &= -\frac{1}{|A_v|} \left( \frac{5 + V_E}{V_T} \right) - \frac{1}{|A_v|} + \left( \frac{5 + V_E}{5} \right) + \frac{V_T}{5} + \frac{1}{|A_v|} \left( \frac{5 + V_E}{V_T} \right) \\
 \boxed{\frac{R_E}{R_C} = \left( \frac{5 + V_E + V_T}{5} \right) - \frac{1}{|A_v|}} & \quad (16)
 \end{aligned}$$

### 4.3.3 Numerical Computation

Setting  $A_v = 30$ :

$$\frac{R_1}{R_C} = \frac{1}{30} - \frac{26 \times 10^{-3}}{5} = 0.02813$$

Found DC value of  $V_E = -4.83V$  from LTSpice.

$$\frac{R_E}{R_C} = \frac{5 - 4.83 + 0.026}{5} - \frac{1}{30} = 0.005867$$

With  $R_C = 15 \text{ k}\Omega$ :

$$R_1 = 0.02813 \times 15 \times 10^3 = 422 \Omega \quad (17)$$

$$R_E = 0.005867 \times 15 \times 10^3 = 88 \Omega \quad (18)$$

The expected values from the calculations were:

$$\boxed{R_C = 15 \text{ k}\Omega, \quad R_E = 88 \Omega, \quad R_1 = 422 \Omega} \quad (19)$$

*Note:* These values were adjusted through iterative optimization in simulation to account for non-ideal effects. The following are the exact resistance values used in the LTSpice simulation. Using these values, the gain of this stage was obtained as  $A_v \approx 22.5$ .

$$\boxed{R_C = 15.2 \text{ k}\Omega, \quad R_E = 225 \Omega, \quad R_1 = 430 \Omega}$$

## 4.4 Input and Output Impedance

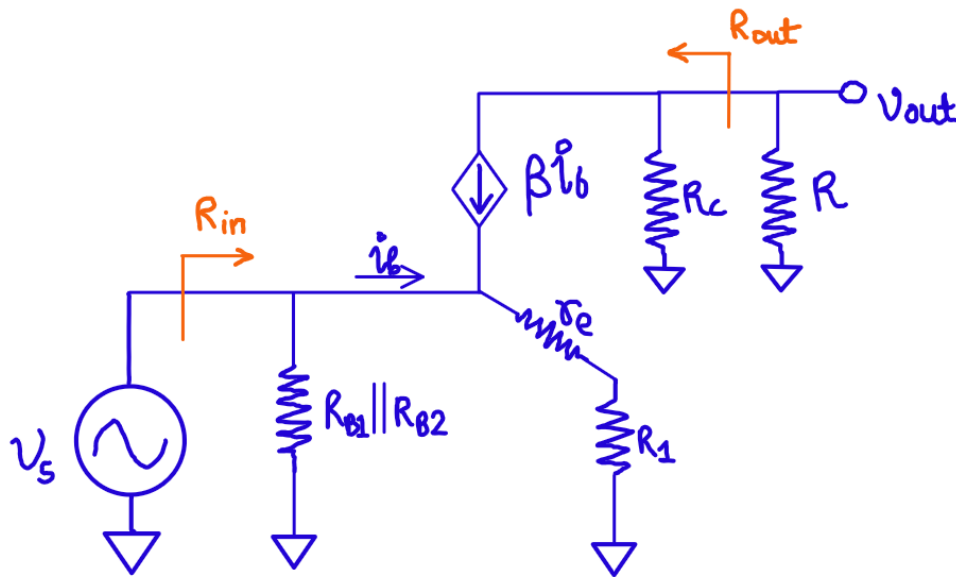


Figure 15: Input and Output impedances of CE Amplifier.

### 4.4.1 Input Impedance

Using the method of impedance reflection:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel \beta(r_e + R_1)$$

Substituting values:

$$r_e = \frac{V_T}{I_E} = \frac{V_T \cdot R_C}{5} = \frac{0.026 \times 15000}{5} = 78 \Omega$$

$$\beta(r_e + R_1) = 200 \times (78 + 430) = 200 \times 508 = 101.6 \text{ k}\Omega$$

$$R_{in} = (113.5 \text{ k}\Omega \parallel 9.5 \text{ k}\Omega \parallel 101.6 \text{ k}\Omega)$$

$$\boxed{R_{in} \approx 8.07 \text{ k}\Omega} \quad (20)$$

*Simulation value:*  $R_{in} = 8.77 \text{ k}\Omega$  (close agreement with calculated value).

#### 4.4.2 Output Impedance

$$R_{out} = R_C \parallel r_o$$

Since  $r_o$  (Early voltage resistance) is very large:

$$\boxed{R_{out} \approx R_C = 15.25 \text{ k}\Omega} \quad (21)$$

### 4.5 Maximum Output Swing

#### 4.5.1 Theory

The output voltage of a CE amplifier is the collector voltage  $V_C$ . This voltage is physically constrained between two limits:

1. **Upper Swing Limit:** If  $V_C$  tries to go above  $V_{CC}$ , the transistor cuts off. The output clips at  $V_{CC}$ .
2. **Lower Swing Limit:**  $V_{C,\min} = V_E + V_{CE,\text{sat}}$ . If  $V_C$  drops too low and approaches  $V_E$ , the transistor enters saturation ( $V_{CE,\text{sat}} \approx 0.2 \text{ V}$ ).

#### 4.5.2 Design for Maximum Swing

By setting  $V_E = -4.94 \text{ V}$ , we push the lower swing limit as low as possible. This creates the *largest possible swing* for  $V_C = V_{out}$  to move, which **maximizes output swing**.

To achieve a symmetrical maximum swing, the DC quiescent collector voltage  $V_{CQ}$  has been biased at the midpoint of the output range. Here,  $V_{CQ} \approx 0 \text{ V}$  (midpoint of the rails).

#### 4.5.3 Calculation

$$V_{o,\min} = V_E + V_{CE,\text{sat}} = -4.94 + 0.2 = -4.74 \text{ V}$$

$$V_{o,\max} = V_{CC} = 5 \text{ V}$$

$$V_{pp,\max} = V_{CC} - (V_E + V_{CE,\text{sat}}) = 5 - (-4.94 + 0.2) = 9.74 \text{ V}$$

$$\boxed{V_{pp,\max} = 9.74 \text{ V}, \quad V_{o,\min} = -4.74 \text{ V}, \quad V_{o,\max} = 5 \text{ V}} \quad (22)$$

#### 4.5.4 Input Signal Limits

The maximum input signal that can be applied without clipping the output:

$$V_{in,pp,\max} = \frac{V_{out,pp,\max}}{|A_v|} = \frac{9.74}{25} \approx 389.6 \text{ mV}_{pp}$$

## 4.6 Integration of First Two Stages

The differential amplifier (gain  $\approx 18$ ) drives the CE amplifier (gain  $\approx 25$ ). The combined gain of the first two stages:

$$A_{v,\text{total}} = A_d \times A_{v,CE} = 21.75 \times 22.56 = 490.68 \quad (\approx 452.18 \text{ in simulation})$$

**Loading considerations:** The output impedance of the differential stage ( $R_{out} \approx 503\Omega$ ) is by design, much smaller than the input impedance of the CE stage ( $R_{in} \approx 8.77\text{ k}\Omega$ ) which allows for minimal signal loss. We use a coupling capacitor  $C_1$  for blocking the DC components of the input signal.

**Capacitances used in the CE-Amp and Coupling capacitor for integration:**

- **Input coupling capacitor:** Capacitor  $C_1$  forms a high pass filter with the input impedance of CE amplifier stage. We keep  $C_1 = 330\mu\text{F}$ , so that this capacitor basically acts only as a DC block and filtering is handled by the Band-Pass stage.
- **Emitter Bypass Capacitor:** Capacitor  $C_3$  allows better DC bias control ( $R_E$  sets DC bias) and increases AC gain by bypassing the  $R_E$  resistor.

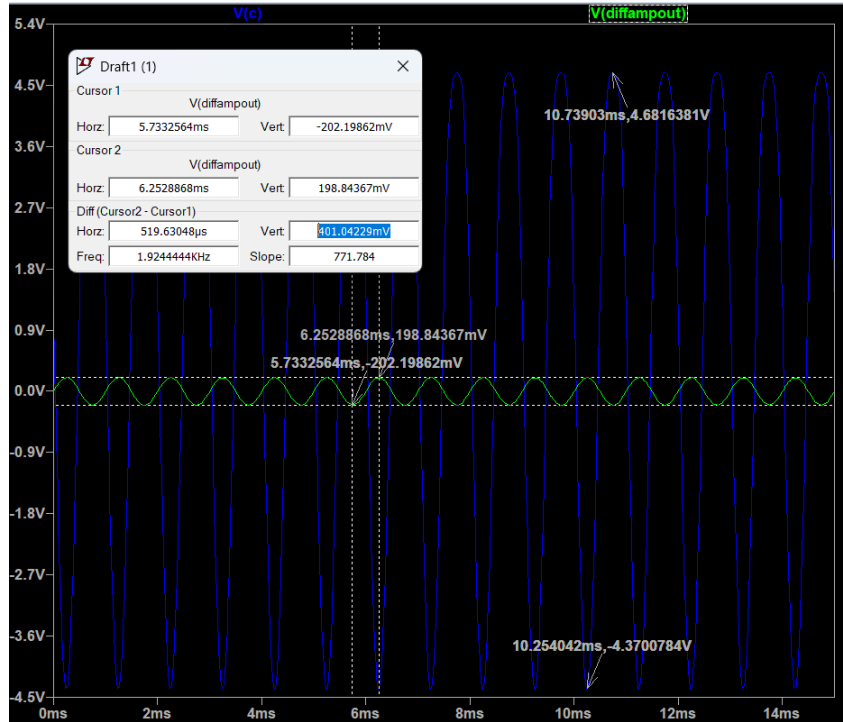


Figure 16: LTSpice output of CE amplifier stage

**Legend:** — Input ( $V_{in1}$ ,  $401\text{mV}_{p-p}$ ) — Output ( $V_{o1}$ ,  $9.05\text{V}_{p-p}$ )

$$\text{Gain till CE Amplifier stage, LTSpice} = \frac{9.05 V_{p-p}}{20 \text{ mV}_{p-p}} = 452.5$$

Gain contribution from this stage:

$$\text{Voltage Gain of CE amp stage } |A_v| = \frac{9.05V}{401.04\text{mV}_{p-p}} = 22.56$$

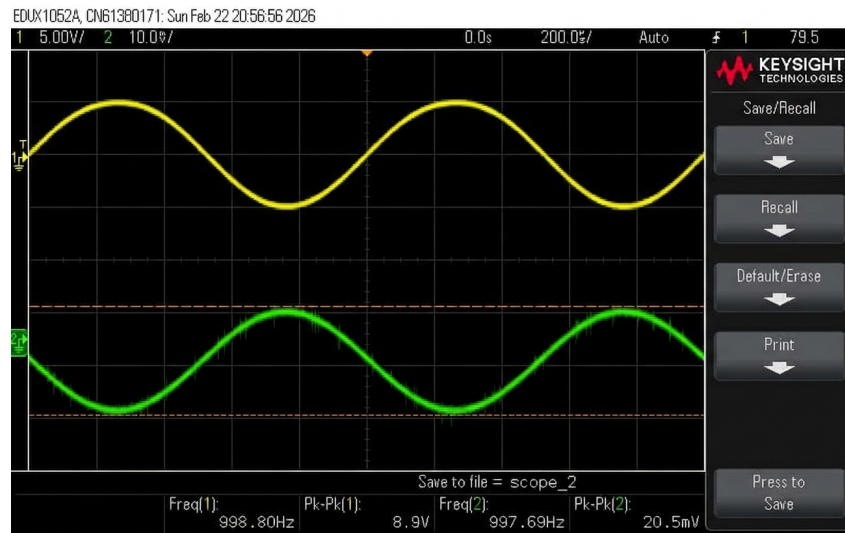


Figure 17: Scope output of hardware implementation: CE Amplifier and Differential pair.

**Legend:** — Input ( $V_{in1}$ ) — Output ( $V_{o1}$ )

$$\text{Gain till CE Amplifier stage, Hardware} = \frac{8.9 V_{p-p}}{20 mV_{p-p}} = 445$$

## 5 Active Band-Pass Filter Stage (20 Hz – 20 kHz)

### 5.1 Theory and Overview

We design an active Bandpass filter to allow only frequencies in the audible range (20Hz-20KHz) while attenuating other frequencies. To achieve this, we have implemented a bandpass filter with a low cutoff frequency at 10 Hz and a high cutoff frequency at 40 kHz.

#### 5.1.1 Band-Pass Implementation

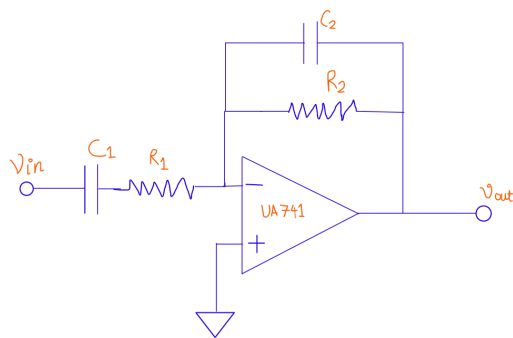


Figure 18: Band Pass Filter Circuit

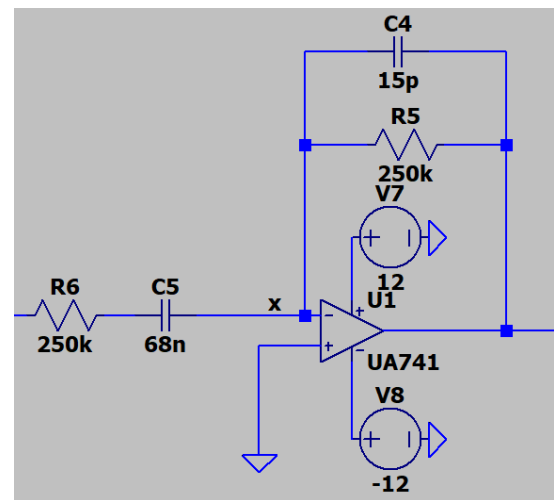


Figure 19: LTSpice BPF Circuit

The band-pass filter is implemented using an **inverting op-amp configuration** with following impedances:

- **Input impedance  $Z_i$ :** A series combination of  $R_1$  and  $C_1$  (forms a high-pass section with cutoff  $\omega_L = 1/(R_1 C_1)$ ).
- **Feedback impedance  $Z_f$ :** A parallel combination of  $R_2$  and  $C_2$  (forms a low-pass section with cutoff  $\omega_H = 1/(R_2 C_2)$ ).

This creates a band-pass response: the circuit passes frequencies between  $\omega_L$  and  $\omega_H$  while attenuating frequencies outside this range.

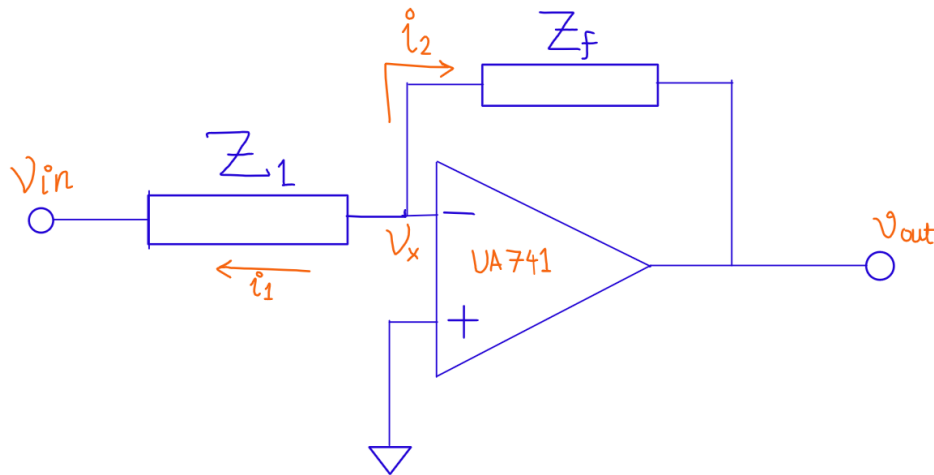


Figure 20: Inverting Configuration

## 5.2 Transfer Function Derivation

*Reference:* Band Pass Filter Circuit

### 5.2.1 Virtual Ground Analysis

Due to the virtual short in the inverting op-amp configuration:

$$V_X = 0 \text{ V} \quad (1)$$

Applying KCL at the inverting input node:

$$i_2 - i_1 = 0 \implies \frac{V_X - V_i}{Z_i} + \frac{V_X - V_o}{Z_f} = 0$$

Substituting  $V_X = 0$ :

$$\frac{-V_i}{Z_i} + \frac{-V_o}{Z_f} = 0$$

$$\boxed{\frac{V_o}{V_i} = -\frac{Z_f}{Z_i}} \quad (23)$$

### 5.2.2 Substituting Complex Impedances

The feedback impedance (parallel  $R_2$  and  $C_2$ ):

$$Z_f = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sR_2C_2}$$

The input impedance (series  $R_1$  and  $C_1$ ):

$$Z_i = R_1 + \frac{1}{sC_1} = \frac{sR_1C_1 + 1}{sC_1}$$

Therefore:

$$\frac{V_o(s)}{V_i(s)} = -\frac{R_2/(1 + sR_2C_2)}{(sR_1C_1 + 1)/(sC_1)} = -\frac{sR_2C_1}{(1 + sR_2C_2)(1 + sR_1C_1)}$$

Multiplying numerator and denominator:

$$H(s) = \frac{V_o(s)}{V_i(s)} = -\frac{sR_2C_1}{(1 + sR_2C_2)(1 + sR_1C_1)} \quad (24)$$

### 5.3 Frequency Response Analysis

#### 5.3.1 Frequency Domain Transfer Function

Substituting  $s = j\omega$ :

$$H(j\omega) = \frac{-j\omega R_2C_1}{(1 + j\omega R_2C_2)(1 + j\omega R_1C_1)}$$

Expanding the denominator:

$$H(j\omega) = \frac{-j\omega R_2C_1}{1 + j\omega(R_1C_1 + R_2C_2) - \omega^2 R_1C_1R_2C_2}$$

#### 5.3.2 Resonant Frequency

The magnitude  $|H(j\omega)|$  is maximized at resonant frequency i.e.  $\omega = \omega_{res}$ :

$$\omega_{res} = \frac{1}{\sqrt{R_1C_1R_2C_2}} \quad (25)$$

Cutoff frequencies from transfer function are:

$$\omega_L = \frac{1}{R_1C_1}, \quad \omega_H = \frac{1}{R_2C_2}$$

The resonant frequency is the geometric mean:

$$\omega_{res} = \sqrt{\omega_L \cdot \omega_H} \quad (26)$$

#### 5.3.3 Magnitude at Resonance

At  $\omega = \omega_{res}$ :

$$\begin{aligned} |H(j\omega)|_{\omega=\omega_{res}} &= \frac{R_2C_1}{R_1C_1 + R_2C_2} \\ &= \frac{R_2}{R_1} \cdot \frac{C_1}{C_1 + C_2 \cdot \frac{R_2}{R_1}} \end{aligned}$$

**Design choice:**  $C_1 \gg C_2$ . Under this condition:

$$|H(j\omega)|_{\omega=\omega_{res}} \approx \frac{R_2}{R_1}$$

To ensure the gain is  $\leq 1$  everywhere in the passband:

$$R_1 = R_2 \quad (27)$$

This ensures **unity passband gain** (0 dB gain), so the filter stage does not add any voltage amplification.

## 5.4 Cutoff Frequency Design

### 5.4.1 Resistor Selection

We set  $R_1 = R_2 = 250 \text{ k}\Omega$ .

**Rationale:** The input impedance of the filter stage must be much greater than the output impedance of the CE amplifier ( $R_{out,CE} = 15.25 \text{ k}\Omega$ ) to prevent loading:

$$R_1 = 250 \text{ k}\Omega \gg 15.25 \text{ k}\Omega$$

Thus, we've successfully evaded the possibility of Loading of CE amplifier output.

### 5.4.2 Lower Cutoff Frequency ( $\omega_L = 10 \text{ Hz}$ )

*Note: The lower cutoff frequency is set to 10 Hz (below the 20 Hz audio range) to ensure the 20 Hz signal is in the passband with minimal attenuation.*

$$\begin{aligned}\omega_L &= \frac{1}{R_1 C_1} = 2\pi \times 10 \\ C_1 &= \frac{1}{\omega_L R_1} = \frac{1}{2\pi \times 10 \times 250 \times 10^3} \\ C_1 &= 63.66 \text{ nF}\end{aligned}\tag{28}$$

We used standard value  $C_1 = 68 \text{ nF}$ .

### 5.4.3 Upper Cutoff Frequency ( $\omega_H = 40 \text{ kHz}$ )

*Note: The upper cutoff frequency is set to 40 kHz (above the 20 kHz audio range) to ensure minimal attenuation at 20 kHz.*

$$\begin{aligned}\omega_H &= \frac{1}{R_2 C_2} = 2\pi \times 40 \times 10^3 \\ C_2 &= \frac{1}{\omega_H R_2} = \frac{1}{2\pi \times 40 \times 10^3 \times 250 \times 10^3} \\ C_2 &= 15.9 \text{ pF}\end{aligned}\tag{29}$$

We used standard value  $C_1 = 15 \text{ pF}$ .

### 5.4.4 Chosen Component Values

After selecting standard component values:

- $f_L = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi \times 250 \text{ k} \times 68 \text{ nF}} = 42.44 \text{ kHz}$
- $f_H = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi \times 250 \text{ k} \times 15 \text{ pF}} = 9.36 \text{ Hz}$

## 5.5 AC Response and Bode Plot Analysis

The Bode plot of the band-pass filter  $H(j\omega)$  has the following characteristics:

- **Below  $\omega_L$ :** The gain increases at +20 dB/decade (high-pass behavior from the  $sR_1C_1$  term).
- **Between  $\omega_L$  and  $\omega_H$  (passband):** The gain is approximately constant at 0 dB (since  $R_1 = R_2$ ).
- **Above  $\omega_H$ :** The gain decreases at -20 dB/decade (low-pass behavior from the  $R_2C_2$  feedback).

In the passband, phase is approximately  $-180^\circ$  (inverting amplifier).

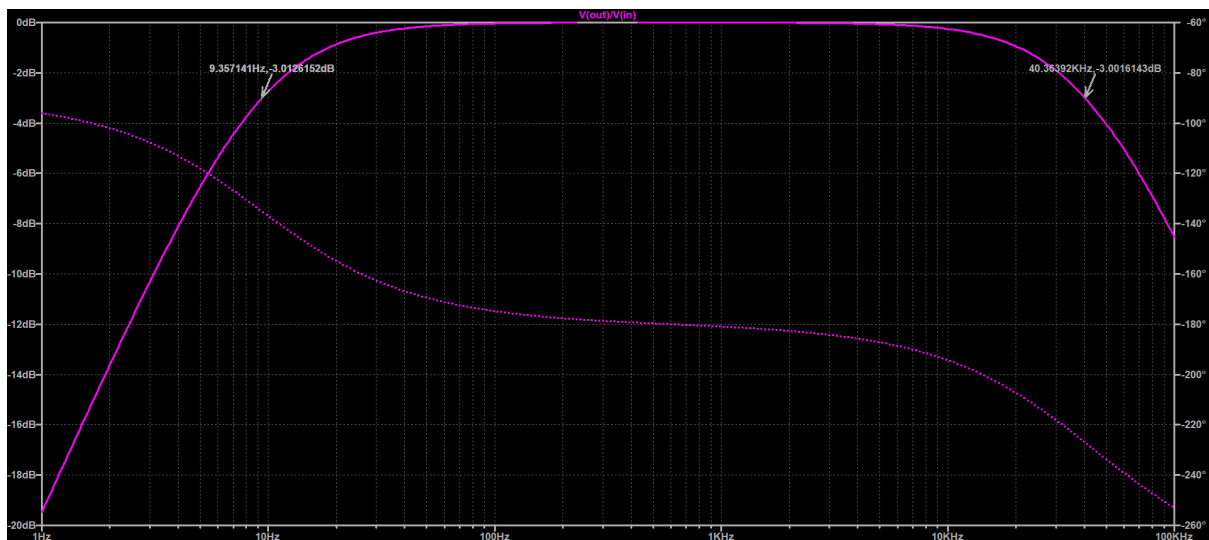


Figure 21: LTSpice Bode Plot Filter stage

**Legend:** — Magnitude — Phase

**Lower Cutoff Frequency (-3dB Cutoff)  $f_L$ :** 9.357141 Hz, at, -3.0126152 dB

**Upper Cutoff Frequency (-3dB Cutoff)  $f_H$ :** 40.36392 kHz, at -3.0016143 dB

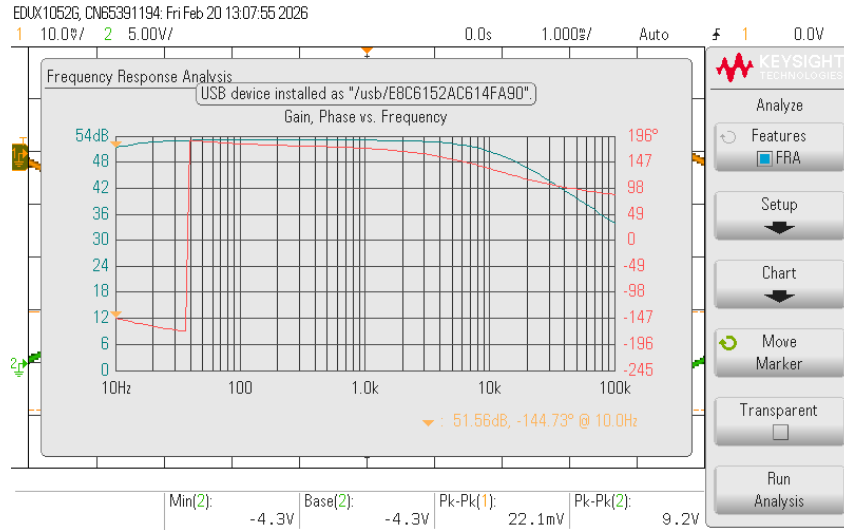


Figure 22: Bode Plot of Output till Filter stage

**Legend:** — Magnitude — Phase

## 5.6 Input and Output Impedance

### 5.6.1 Output Impedance

Assuming an ideal op-amp (output impedance  $\approx 0\Omega$ ):

$$\boxed{R_{out} = 0\Omega} \quad (30)$$

This feature of low output impedance allows this stage to drive the subsequent power amplifier stage without loading. This is one of the key reasons we chose Active filters over Passive Filters.

### 5.6.2 Input Impedance

Looking into the circuit from the input terminal:

$$R_{in} = \frac{1}{j\omega C_1} + R_1$$

$$|R_{in}| = \sqrt{\frac{1}{\omega^2 C_1^2} + R_1^2} \approx R_1 \quad (\text{for } \omega \gg 1/(R_1 C_1))$$

$$\boxed{R_{in} \approx 290\text{ k}\Omega} \quad (31)$$

This is sufficiently large to prevent loading of the CE amplifier stage.

## 6 Power Amplifier Stage (Class AB)

### 6.1 Theory

#### 6.1.1 Need for a Power Amplifier Stage

The preceding stages (differential amplifier, CE stage, filter) provide voltage gain but cannot deliver significant current to a low-impedance load such as a loudspeaker (typically  $4\ \Omega$  to  $16\ \Omega$ ). The power amplifier stage provides:

- High current gain (current buffering).
- Unity voltage gain - the signal amplitude is preserved.
- Low output impedance to drive the speaker.

#### 6.1.2 Why Class AB?

Different amplifier classes have different tradeoffs:

| Class | Efficiency           | Linearity | Crossover Distortion |
|-------|----------------------|-----------|----------------------|
| A     | Low ( $\sim 25\%$ )  | Excellent | None                 |
| B     | High ( $\sim 78\%$ ) | Poor      | Severe               |
| AB    | Moderate (50–70%)    | Good      | Minimal              |

Table 1: Comparison of amplifier classes.

**Class AB** is the optimal choice for audio amplification because:

- It eliminates the crossover distortion inherent in Class B by keeping both output transistors slightly on near the zero-crossing region.
- It provides much better efficiency than Class A.
- It maintains good signal fidelity across the audio range.

#### 6.1.3 Push-Pull Configuration

A Class AB push-pull stage uses a complementary pair of transistors:

- An **NPN transistor** handles the positive half-cycle of the output.
- A **PNP transistor** handles the negative half-cycle.
- A small bias current ( $I_{\text{bias}}$ ) flows through both transistors at the quiescent point, ensuring both are slightly on to avoid crossover distortion.

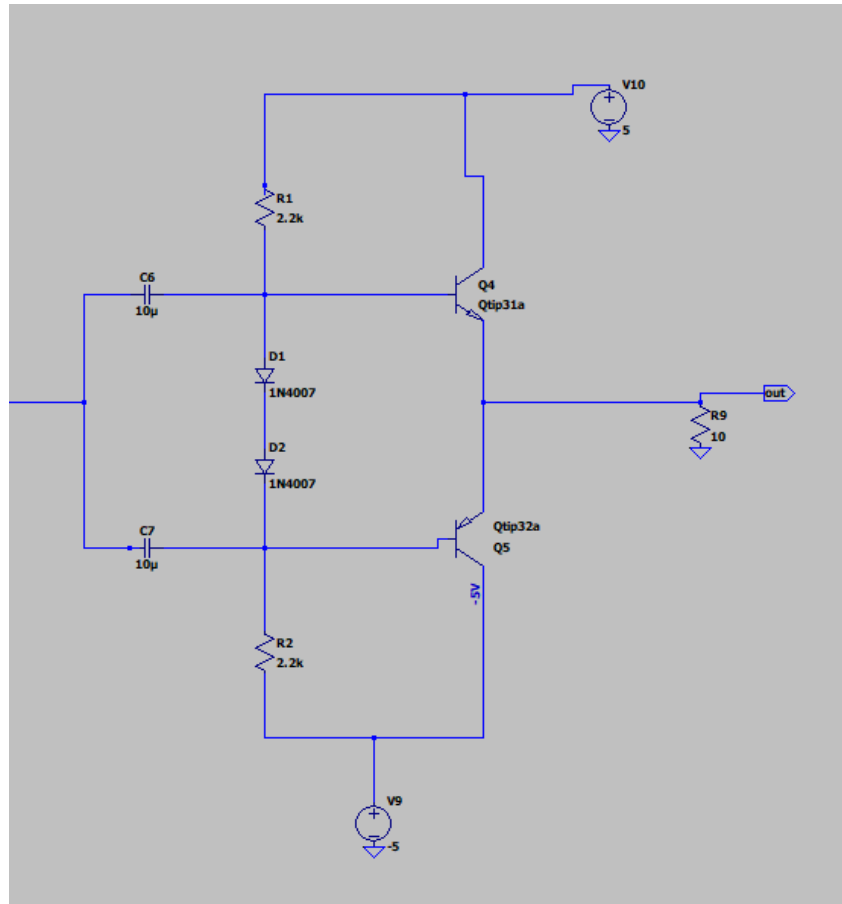


Figure 23: Class AB push-pull output stage circuit.

## 6.2 Circuit Configuration and Biasing

### 6.2.1 Biasing for Class AB Operation

The bias voltage between the bases of the NPN and PNP transistors is set to approximately:

$$V_{bias} \approx 2V_{BE} \approx 1.4 \text{ V}$$

This is achieved using two silicon diodes placed between the transistor bases. To maintain proper diode conduction, a bias current of approximately:

$$I_{bias} \approx 2 \text{ mA}$$

is selected.

The total supply voltage across the bias network is:

$$V_{total} = V_{CC} - V_{EE} = 5 - (-5) = 10 \text{ V}$$

Voltage across bias resistors:

$$V_R = V_{total} - V_{bias}$$

$$V_R = 10 - 1.4 = 8.6 \text{ V}$$

Thus,

$$R_1 + R_2 = \frac{V_R}{I_{bias}}$$

$$R_1 + R_2 = \frac{8.6}{0.002}$$

$$R_1 + R_2 = 4.3 \text{ k}\Omega$$

Choosing symmetrical biasing:

$$R_1 = R_2 \approx 2.2 \text{ k}\Omega$$

This maintains a stable bias voltage of approximately 1.4 V between the transistor bases.

### 6.2.2 Coupling Capacitor Design

The capacitors  $C_6$  and  $C_7$  act as coupling capacitors that block DC while allowing the AC signal to pass to the Class AB output stage. These capacitors form a high-pass RC network with the input resistance of the bias network.

The cutoff frequency of a high-pass RC circuit is given by

$$f_c = \frac{1}{2\pi R_{in} C}$$

where  $R_{in}$  is the equivalent input resistance seen by the capacitor and  $C$  is the coupling capacitance.

Rearranging the equation to obtain the required capacitance,

$$C = \frac{1}{2\pi R_{in} f_c}$$

In this circuit, the input resistance is determined by the bias resistors  $R_7$  and  $R_8$ :

$$R_{in} = R_1 \parallel R_2$$

$$R_{in} = \frac{R_1 R_2}{R_1 + R_2}$$

Substituting  $R_7 = R_8 = 2.2 \text{ k}\Omega$ ,

$$R_{in} = \frac{2.2 \text{ k}\Omega}{2} = 1.1 \text{ k}\Omega$$

Assuming a lower cutoff frequency of  $f_c = 20 \text{ Hz}$  for audio applications,

$$C = \frac{1}{2\pi(1100)(20)}$$

$$C \approx 7.23 \mu\text{F}$$

Thus, the nearest standard capacitor value is chosen as

$$C \approx 10 \mu\text{F}$$

which ensures proper low-frequency coupling without significant signal attenuation.

## 6.3 Design Derivation and Component Values

### 6.3.1 Output Transistor Selection

We have used complementary transistor pair **Tip31C** (NPN) and **Tip32C** (PNP).

Key parameters:

- Maximum collector current:  $I_C^{max} \approx 3\text{ A}$
- Current gain:  $h_{FE} \approx 50$
- Maximum collector-emitter voltage:  $V_{CE(max)} \approx 100\text{ V}$

These ratings comfortably support our requirements.

### 6.3.2 Component Value Justification

- Bias resistors:  $R_1 = R_2 \approx 2.2\text{ k}\Omega$
- Bias diodes: Two silicon diodes providing  $\approx 1.4\text{ V}$
- Load resistor:  $R_L = 10\text{ }\Omega$

## 6.4 Unity Voltage Gain Verification

The output stage operates as a **common-collector (emitter follower)** amplifier.

$$A_v = \frac{R_L}{r_e + R_L}$$

where

$$r_e = \frac{V_T}{I_{CQ}}$$

Assuming a quiescent current:

$$I_{CQ} = 50\text{ mA}$$

$$r_e = \frac{25\text{ mV}}{50\text{ mA}}$$

$$r_e = 0.5\text{ }\Omega$$

Thus,

$$A_v = \frac{10}{10 + 0.5}$$

$$A_v = 0.95 \approx 1$$

Therefore, the power stage provides **unity voltage gain**, satisfying the design specification.

## 6.5 Power Output Calculation

### 6.5.1 Theory

Maximum RMS output voltage required for 1.5 W power:

$$P = \frac{V_{rms}^2}{R_L}$$

$$V_{rms} = \sqrt{PR_L}$$

$$V_{rms} = \sqrt{1.5 \times 10}$$

$$V_{rms} = 3.87 V$$

Peak output voltage:

$$V_{peak} = \sqrt{2}V_{rms}$$

$$V_{peak} = 5.47 V$$

Peak output current:

$$I_{peak} = \frac{V_{peak}}{R_L}$$

$$I_{peak} = \frac{5.47}{10}$$

$$I_{peak} = 0.547 A$$

Thus, each output transistor must handle approximately:

$$I_C \approx 0.55 A$$

### 6.5.2 Efficiency

For Class AB amplifiers, the theoretical maximum efficiency approaches:

$$\eta_{max} \approx 78.5\%$$

Assuming ideal operation:

$$P_{supply} \approx \frac{P_{out}}{\eta}$$

$$P_{supply} \approx \frac{1.5}{0.785}$$

$$P_{supply} \approx 1.91 W$$

Power dissipated in transistors:

$$P_{diss} = P_{supply} - P_{out}$$

$$P_{diss} \approx 0.41 \text{ W}$$

## 6.6 Input and Output Impedance

The power stage uses a complementary transistor pair: TIP31C (NPN) and TIP32C (PNP).

Assuming a quiescent current:

$$I_{CQ} = 50 \text{ mA}$$

The small-signal emitter resistance is

$$r_e = \frac{V_T}{I_{CQ}}$$

where  $V_T \approx 25 \text{ mV}$ .

$$r_e = \frac{25 \text{ mV}}{50 \text{ mA}}$$

$$r_e = 0.5 \Omega$$

### 6.6.1 Output Impedance

For an emitter follower stage:

$$R_{out} = r_e + \frac{R_S}{\beta + 1}$$

Since the preceding stage has very low source impedance,

$$R_S \approx 0$$

Thus,

$$R_{out} \approx r_e$$

$$R_{out} \approx 0.5 \Omega$$

This low output impedance allows efficient power transfer to the  $10 \Omega$  load.

### 6.6.2 Input Impedance

The input impedance of the emitter follower is given by:

$$R_{in} = (\beta + 1)(R_L + r_e)$$

Assuming a transistor current gain of

$$\beta \approx 25$$

$$R_{in} = (25 + 1)(10 + 0.5)$$

$$R_{in} = 26 \times 10.5$$

$$R_{in} \approx 273 \Omega$$

Therefore,

$$R_{in} \approx 270 \Omega$$

This impedance is sufficiently high to avoid loading the preceding gain stage while maintaining strong current drive capability.

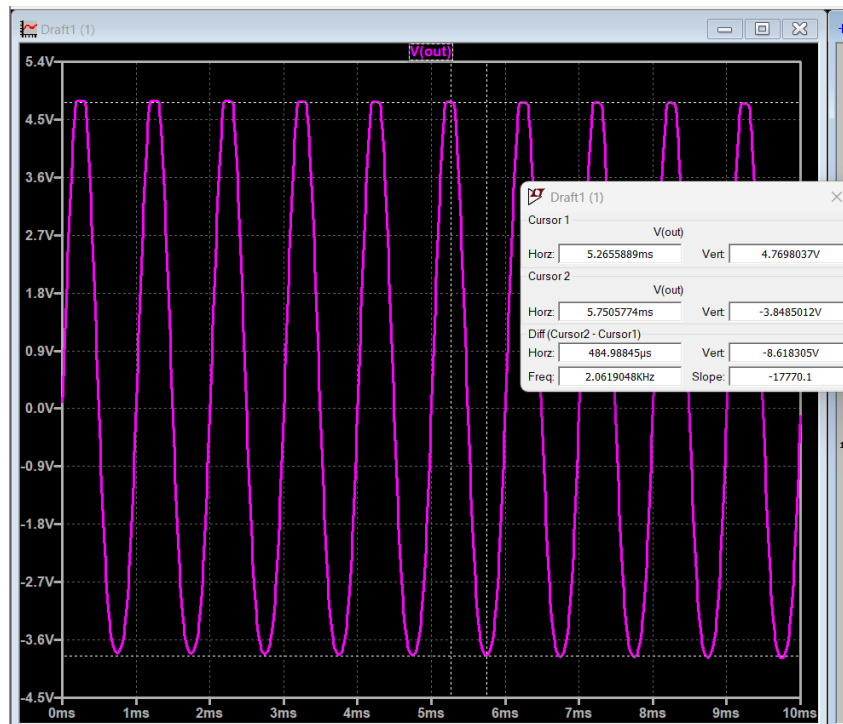


Figure 24: LTSpice Simulation

**Legend:** —  $V_{out}$  for 20mVpp

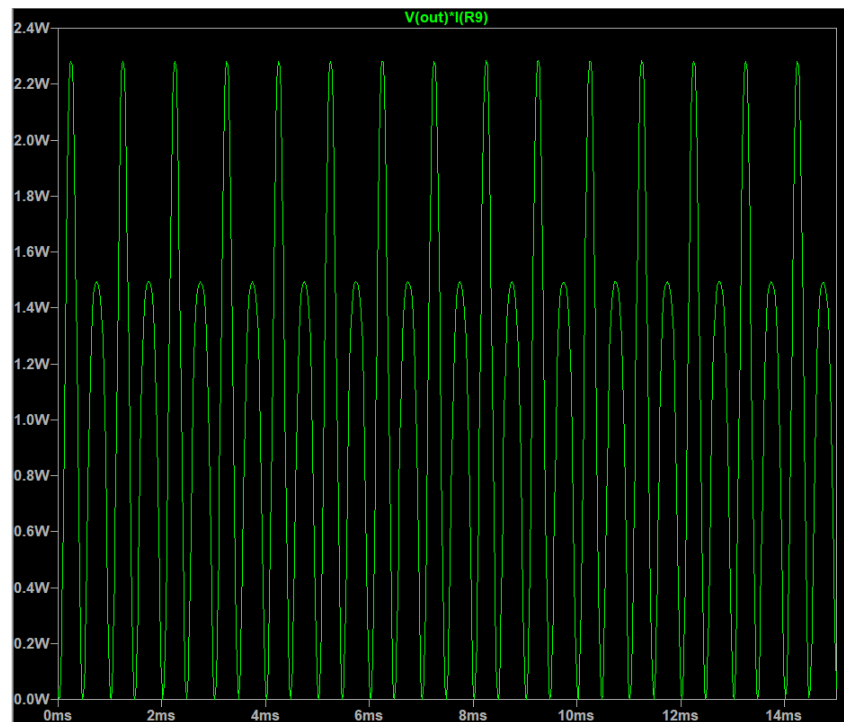


Figure 25: LTSpice Power Amplifier power output

**Legend:** — Power of  $V_{out}$  for 20mVpp, Peak value of power 2.3W > 1.5W

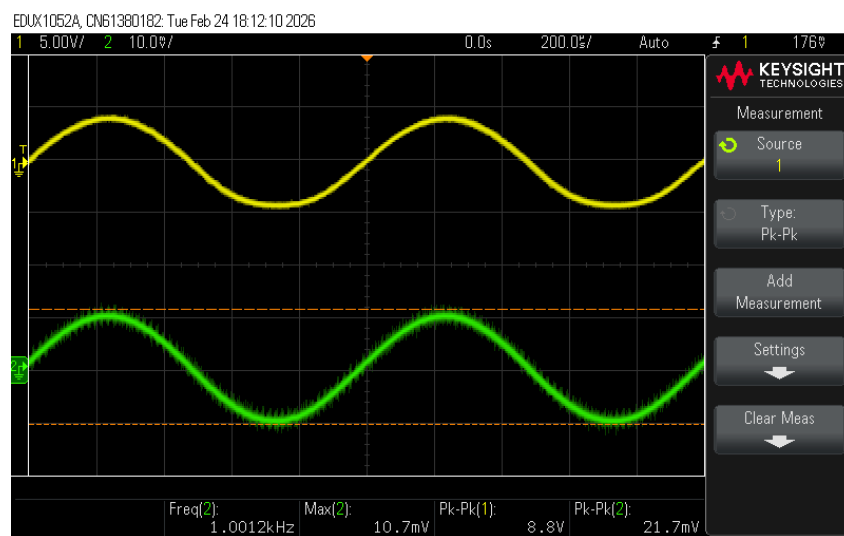


Figure 26: Scope output of hardware implementation: Power Amplifier Output for 20mVpp input

**Legend:** — Output(10Ω load) — Input of Differential Pair

## 7 Overall Performance and Analysis

### 7.1 Complete Amplifier Chain

The complete four-stage amplifier chain is:

$$\text{Input} \rightarrow \text{Diff. Amp.} \xrightarrow{A_d} \text{CE Stage} \xrightarrow{A_v} \text{BPF} \xrightarrow{A_c} \text{Class AB} \xrightarrow{A_p} \text{Speaker}$$

### 7.2 Final Gain Verification

#### 7.2.1 AC Analysis (Bode Plot)

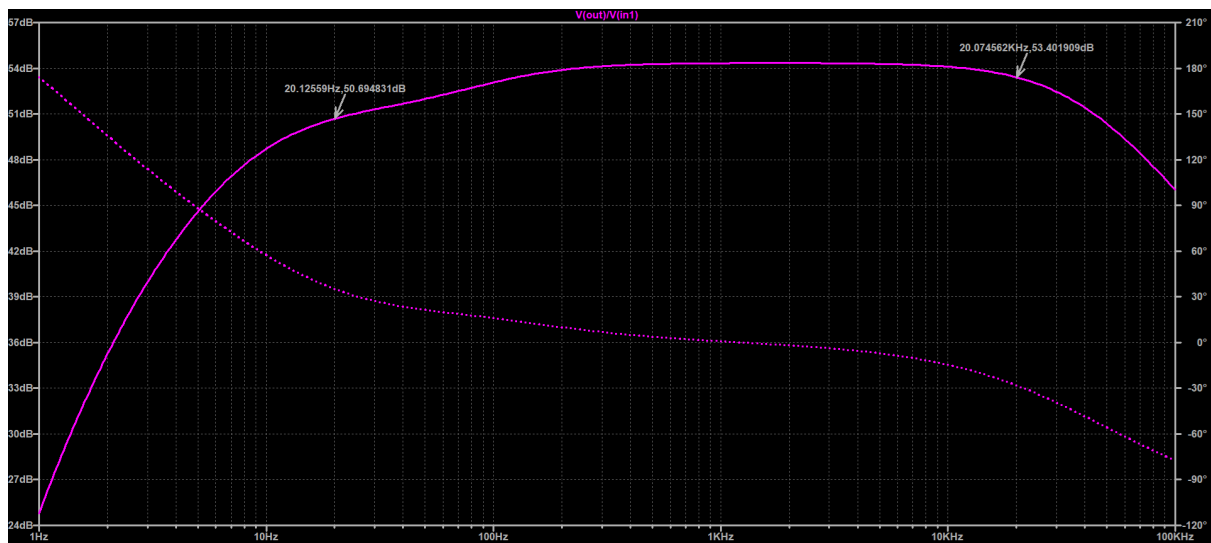


Figure 27: Output AC Analysis (LT Spice)

**Legend:** — Magnitude    - - - Phase Plot

**Lower-End Operating Band Limit:** 20.12559 Hz, 50.694831 dB

**Higher-End Operating Band Limit:** 20.074562 kHz, 53.401909 dB

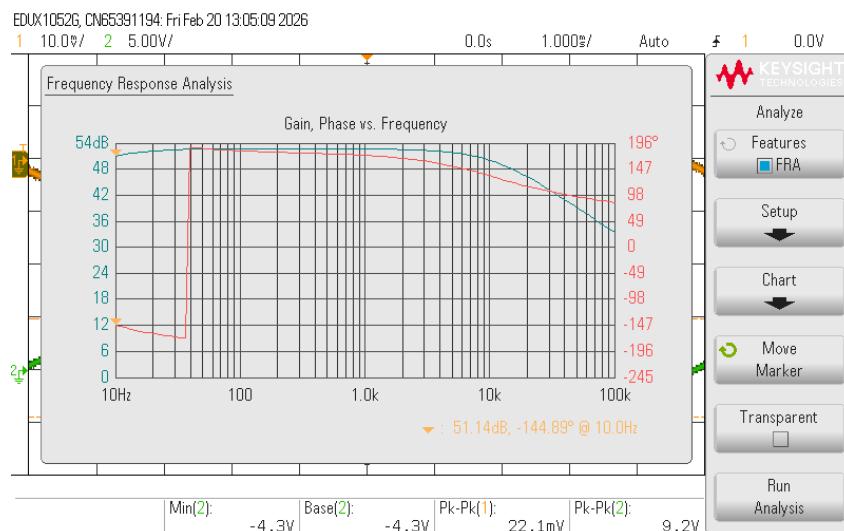


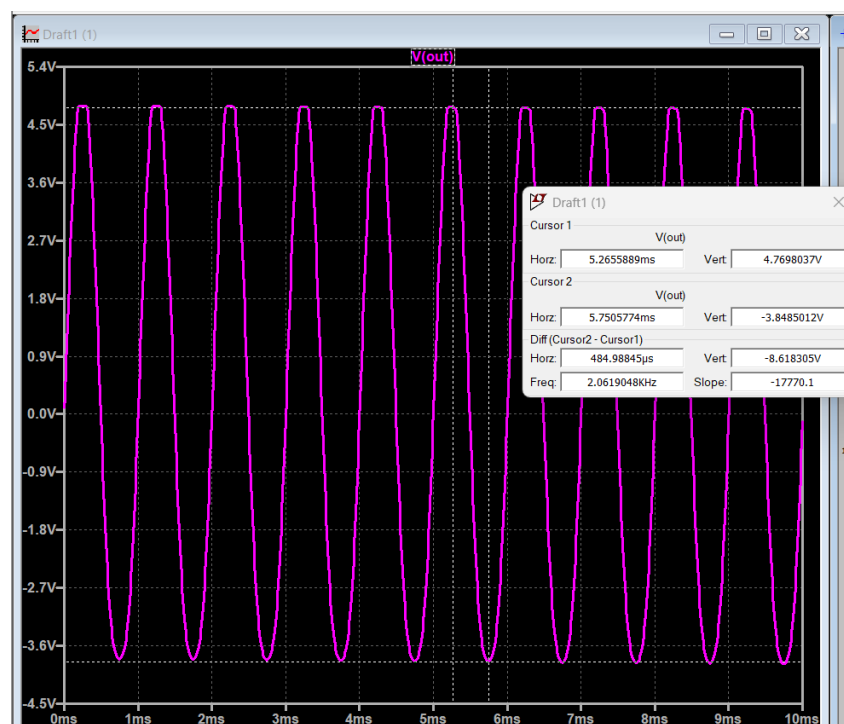
Figure 28: Output AC Analysis (Hardware)

Legend: — Magnitude — Phase

Lower-End Operating Band Limit: 20 Hz, 51.5 dB, (approx)

Higher-End Operating Band Limit: 20 kHz, 48 dB, (approx)

## 7.2.2 Transient Analysis

Figure 29: LTSpice Simulation Overall,  $20\text{mV}_{p-p}\text{input}$ 

Legend: —  $V_{out}$

Overall Output Voltage: 8.63 Vpp

Overall Gain: 431

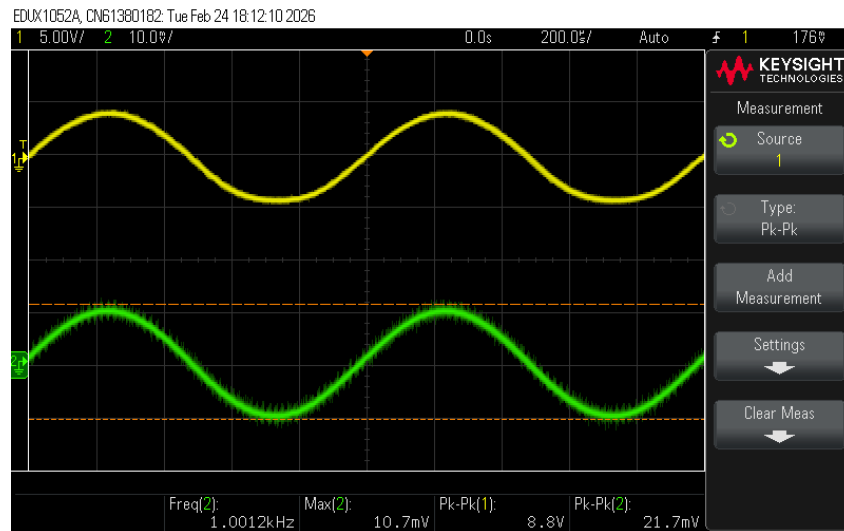


Figure 30: Hardware Overall Output, 20mVp-p input

**Legend:** — Output(10Ω load) — Input of Differential Pair

**Overall Output Voltage:** 8.8 Vpp

**Overall Gain:** 440

## 7.3 Total Harmonic Distortion (THD) Analysis

### 7.3.1 Theory

Total Harmonic Distortion quantifies the nonlinearity introduced by the amplifier. It is defined as:

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1} \times 100\%$$

where  $V_1$  is the amplitude of the fundamental frequency and  $V_2, V_3, \dots$  are the amplitudes of the harmonics.

### 7.3.2 THD Measurement Method Employed(LTSpice)

In **LTSpice**, we measure THD using the built-in Fourier analysis tool.

### 7.3.3 THD for Input Sine Wave of 10 mV pk-pk

| Harmonic Number | Frequency [Hz] | Fourier Component | Normalized Component | Phase [degree] | Normalized Phase [deg] |
|-----------------|----------------|-------------------|----------------------|----------------|------------------------|
| 1               | 1.000e+03      | 2.533e+00         | 1.000e+00            | 89.24°         | 0.00°                  |
| 2               | 2.000e+03      | 1.026e-01         | 4.049e-02            | 178.46°        | 89.22°                 |
| 3               | 3.000e+03      | 1.961e-02         | 7.742e-03            | 89.21°         | -0.03°                 |
| 4               | 4.000e+03      | 3.978e-03         | 1.571e-03            | 0.39°          | -88.85°                |
| 5               | 5.000e+03      | 4.227e-04         | 1.669e-04            | -91.74°        | -180.98°               |
| 6               | 6.000e+03      | 8.432e-05         | 3.329e-05            | 22.27°         | -66.97°                |
| 7               | 7.000e+03      | 2.166e-05         | 8.550e-06            | 20.59°         | -68.65°                |
| 8               | 8.000e+03      | 6.175e-05         | 2.438e-05            | 45.85°         | -43.39°                |
| 9               | 9.000e+03      | 9.220e-05         | 3.640e-05            | 109.87°        | 20.63°                 |

Partial Harmonic Distortion: 4.125390%

Total Harmonic Distortion: 4.125782%

#### 7.3.4 THD for Input Sine Wave of 20 mV pk-pk

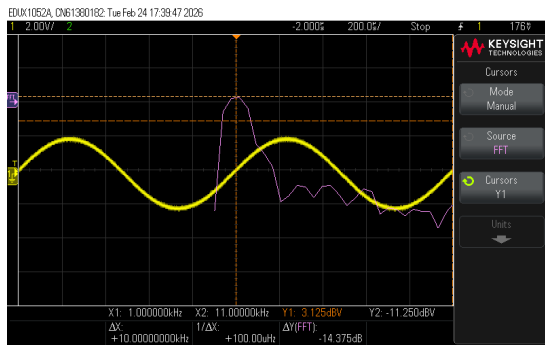
| Harmonic Number | Frequency [Hz] | Fourier Component | Normalized Component | Phase [degree] | Normalized Phase [deg] |
|-----------------|----------------|-------------------|----------------------|----------------|------------------------|
| 1               | 1.000e+03      | 3.435e+00         | 1.000e+00            | 89.28°         | 0.00°                  |
| 2               | 2.000e+03      | 2.317e-01         | 6.746e-02            | 178.58°        | 89.30°                 |
| 3               | 3.000e+03      | 5.956e-02         | 1.734e-02            | 89.50°         | 0.22°                  |
| 4               | 4.000e+03      | 1.472e-02         | 4.285e-03            | 0.66°          | -88.62°                |
| 5               | 5.000e+03      | 2.594e-03         | 7.552e-04            | -90.36°        | -179.64°               |
| 6               | 6.000e+03      | 7.091e-04         | 2.064e-04            | 3.92°          | -85.36°                |
| 7               | 7.000e+03      | 2.477e-04         | 7.213e-05            | -88.13°        | -177.41°               |
| 8               | 8.000e+03      | 2.042e-04         | 5.944e-05            | 24.19°         | -65.09°                |
| 9               | 9.000e+03      | 1.807e-04         | 5.261e-05            | 102.33°        | 13.05°                 |

Partial Harmonic Distortion: 6.978840%

Total Harmonic Distortion: 6.979090%

## 7.4 THD of Hardware

### 7.4.1 Peak Voltage = 10mVpp



(a) Fundamental Frequency(1kHz) in FFT

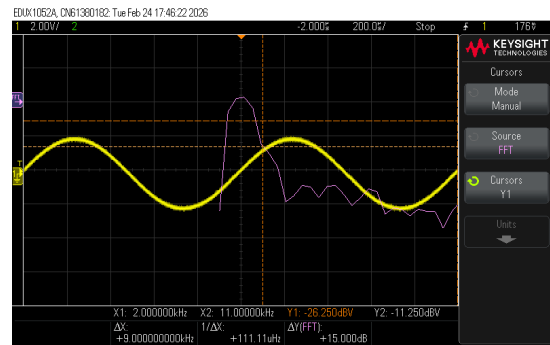
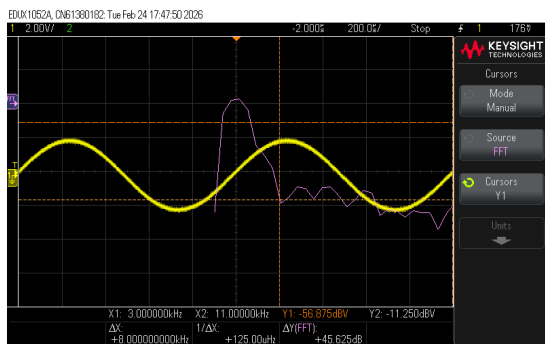
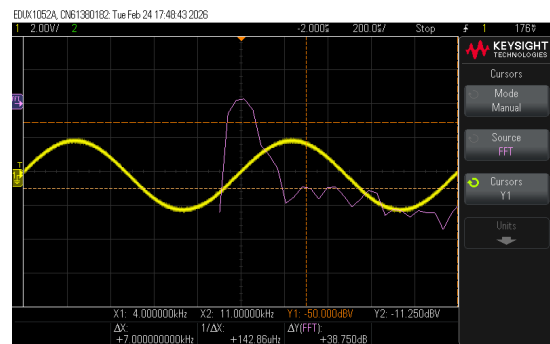
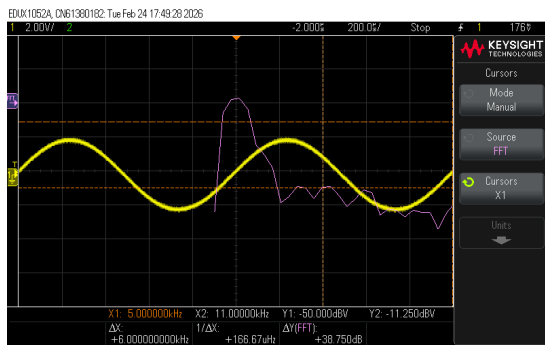
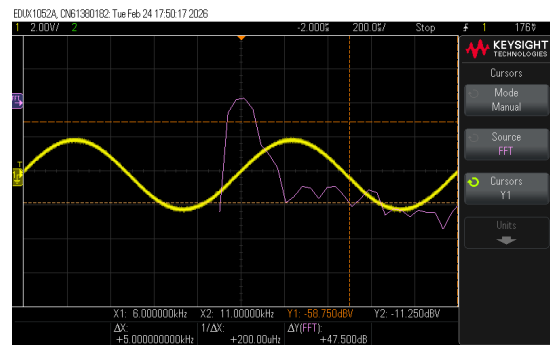
(b) 2<sup>nd</sup> harmonic (2 kHz) in FFT(c) 3<sup>rd</sup> harmonic(3kHz) in FFT(d) 4<sup>th</sup> harmonic(4kHz) in FFT(e) 5<sup>th</sup>harmonic(5kHz) in FFT(f) 6<sup>th</sup>harmonic(6kHz) in FFT

Figure 31: Experimental FFT measurements of output signal harmonics (1 kHz – 6 kHz) for THD calculation

**Legend:** — Output of Circuit (10 $\Omega$  load) — FFT of Output signal

## THD Calculation

- **THD at 1 kHz, nominal input level**

The harmonic amplitudes obtained from the Fourier analysis are

$$V_1 = 3.125 \text{ dBV}$$

$$V_2 = -26.250 \text{ dBV}$$

$$V_3 = -56.875 \text{ dBV}$$

$$V_4 = -50.000 \text{ dBV}$$

$$V_5 = -50.000 \text{ dBV}$$

$$V_6 = -58.750 \text{ dBV}$$

Converting from dBV to volts using

$$V = 10^{(\text{dBV}/20)}$$

$$V_1 \approx 1.43 \text{ V}$$

$$V_2 \approx 0.04870 \text{ V}$$

$$V_3 \approx 0.00143 \text{ V}$$

$$V_4 \approx 0.00316 \text{ V}$$

$$V_5 \approx 0.00316 \text{ V}$$

$$V_6 \approx 0.00115 \text{ V}$$

The Total Harmonic Distortion (THD) is defined as

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + V_5^2 + V_6^2}}{V_1}$$

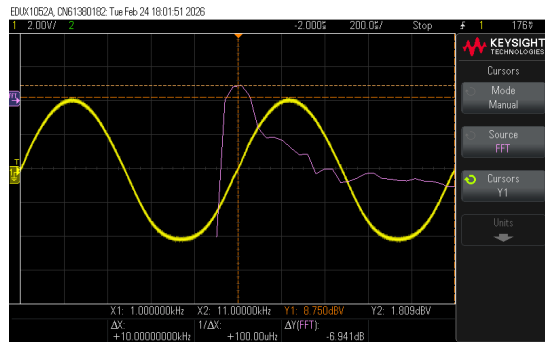
Substituting the values,

$$\sqrt{(0.050)^2 + (0.00143)^2 + (0.00316)^2 + (0.00316)^2 + (0.00115)^2} \approx 0.0502$$

$$\text{THD} = \frac{0.0502}{1.43} \approx 0.03559$$

$$\boxed{\text{THD} \approx 3.56\%}$$

### 7.4.2 Peak Voltage = 20mVpp



(a) Fundamental Frequency(1kHz) in FFT

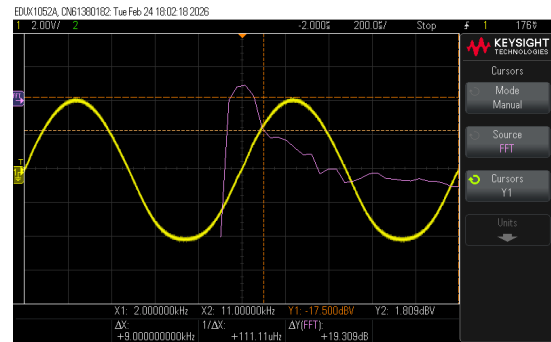
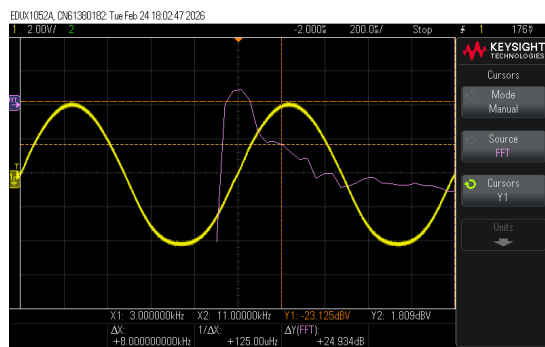
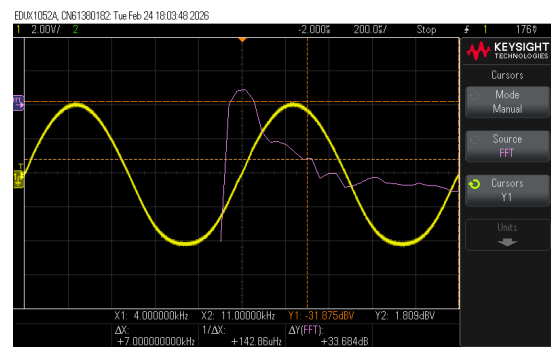
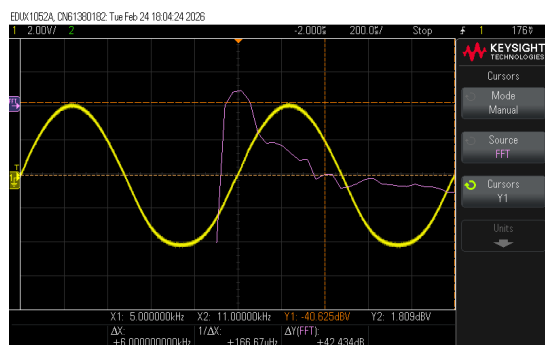
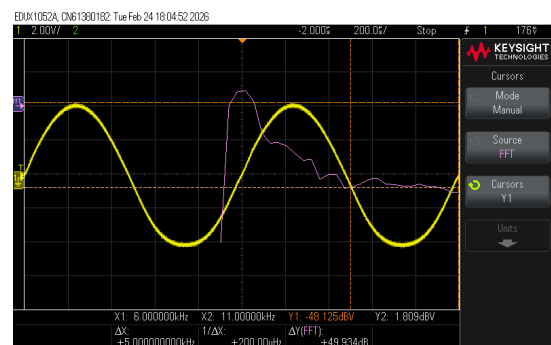
(b) 2<sup>nd</sup> harmonic (2 kHz) in FFT(c) 3<sup>rd</sup> harmonic(3kHz) in FFT(d) 4<sup>th</sup> harmonic(4kHz) in FFT(e) 5<sup>th</sup>harmonic(5kHz) in FFT(f) 6<sup>th</sup>harmonic(6kHz) in FFT

Figure 32: Experimental FFT measurements of output signal harmonics (1 kHz – 6 kHz) for THD calculation

**Legend:** — Output of Circuit (10Ω load) — FFT of Output signal

### THD Calculation

- THD at 1 kHz, higher input level

The harmonic amplitudes obtained from Fourier analysis are

$$\begin{aligned}
V_1 &= 8.750 \text{ dBV} \\
V_2 &= -17.500 \text{ dBV} \\
V_3 &= -23.125 \text{ dBV} \\
V_4 &= -31.875 \text{ dBV} \\
V_5 &= -40.625 \text{ dBV} \\
V_6 &= -48.125 \text{ dBV}
\end{aligned}$$

Converting from dBV to volts using

$$V = 10^{(\text{dBV}/20)}$$

$$\begin{aligned}
V_1 &\approx 2.74 \text{ V} \\
V_2 &\approx 0.133 \text{ V} \\
V_3 &\approx 0.0698 \text{ V} \\
V_4 &\approx 0.0254 \text{ V} \\
V_5 &\approx 0.0093 \text{ V} \\
V_6 &\approx 0.0039 \text{ V}
\end{aligned}$$

The Total Harmonic Distortion (THD) is defined as

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + V_5^2 + V_6^2}}{V_1}$$

Substituting the values,

$$\sqrt{(0.133)^2 + (0.0698)^2 + (0.0254)^2 + (0.0093)^2 + (0.0039)^2} \approx 0.153$$

$$\text{THD} = \frac{0.153}{2.74} \approx 0.0558$$

$$\boxed{\text{THD} \approx 5.6\%}$$

## 7.5 Slew Rate Analysis

### 7.5.1 Theory

The slew rate (SR) defines the maximum rate of change of the output voltage:

$$\text{SR} = \left. \frac{dv_{out}}{dt} \right|_{\max} \quad [\text{V}/\mu\text{s}]$$

The slew rate is typically limited by the internal capacitances of the transistors and op-amp, and the maximum current available to charge/discharge these capacitances:

$$\text{SR} = \frac{I_{\max}}{C_{\text{internal}}}$$

### 7.5.2 LTSpice Simulation Slew Rate

The slew rate of the amplifier was estimated from the transient simulation by observing the slope of the output waveform near the zero-crossing point. At this point the slope of a sinusoidal signal is maximum, making it suitable for estimating the maximum rate of change of the output voltage.

The slew rate is calculated from the slope of the waveform as

$$\text{Slew Rate} = \frac{dV}{dt}$$

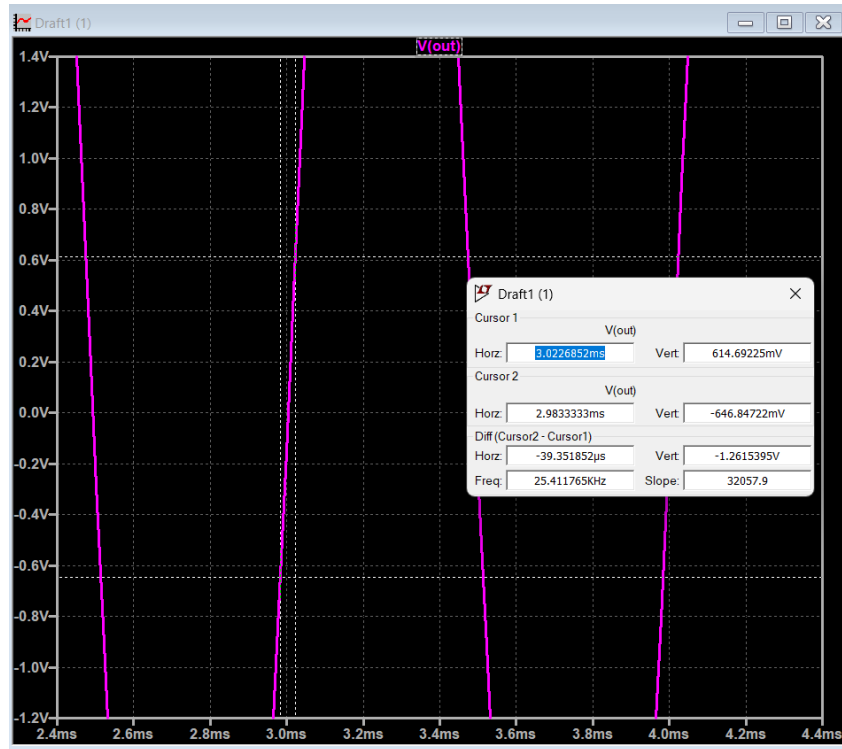


Figure 33: Slew rate measurement from the slope of the output waveform near the zero-crossing point in LTSpice.

From the measured slope of the output waveform in LTSpice, the slew rate is

$$SR \approx 0.0321 \text{ V}/\mu\text{s}.$$

### 7.5.3 Hardware Slew Rate

From the MATLAB-parsed plot of the measured output waveform (plot taken Figure: Hardware Overall Output, 20mVp-p input), the instantaneous rate of change of the signal was analysed. The slope of the waveform was evaluated in the region around the zero-crossing point, where the rate of change of a sinusoidal signal is maximum. The instantaneous values of  $\frac{dV}{dt}$  were obtained from the plot, and the average slope in this region was calculated. This **average value of slope at the waveform crossing** is reported as the slew rate of the hardware amplifier.

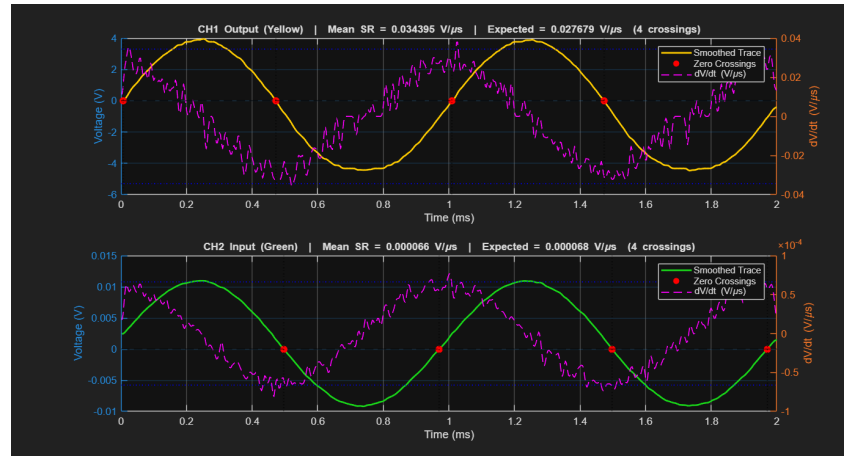


Figure 34: From the our Hardware output plot we extracted instantaneous rate of change of waveform, and average value of the slope at Waveform zero-crossing is reported as Slew rate

Table 2: Slew Rate Analysis Results (Output Channel)

| Parameter        | Value               |
|------------------|---------------------|
| Mean Slew Rate   | 0.034395 V/ $\mu$ s |
| Median Slew Rate | 0.033844 V/ $\mu$ s |

## 8 Conclusion

A four-stage audio amplifier was successfully designed, simulated, and tested:

1. The **BJT differential pair pre-amplifier** provides a differential gain of  $\approx 21.75$  with a CMRR of  $\approx 85$  dB, offering good rejection of common-mode noise.
2. The **CE gain stage** provides the primary voltage amplification (gain  $\approx 22.56$ ) with a maximum output swing of  $9.05 V_{pp}$ , using voltage divider biasing for thermal stability.
3. The **active band-pass filter** confines the signal to the audio band (20 Hz – 20 kHz) with unity passband gain, using an inverting op-amp topology.
4. The **Class AB power amplifier** provides the current drive capability for the loudspeaker load at unity voltage gain, with minimal crossover distortion.

The overall voltage gain of  $\approx 490$  (53.8 dB) was verified through both AC and transient simulations. The amplifier meets the design specifications for gain, bandwidth, and distortion performance.

## 9 Link to Demo on Hardware Circuit

### [Amplifier Demo](#)